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Master's Thesis

KinSim: Development of a long-read simulation tool incorporating kinetic signals

Report submitted in partial fulfillment of the requirements for the degree of:

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WALLONIE-BRUXELLES ENSEIGNEMENT
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To my parents and their never-ending support during my studies, to my brothers, my friends. To my girlfriend who gives me motivation to be better.

Résumé

KinSim: Développement d'un outil de simulation de long reads intégrant les signaux cinétiques

Le séquençage PacBio HiFi produit, en plus de la séquence d'ADN, deux signaux cinétiques par base, l'Inter-Pulse Duration (IPD) et le Pulse Width (PW), dont les variations reflètent les modifications chimiques du brin séquencé, en particulier les méthylations bactériennes m6A, m4C et m5C [1], [2]. Chaque espèce bactérienne possédant un répertoire de méthyltransférases qui lui est propre [3], ces marques constituent une signature spécifique qui pourrait améliorer le *binning* métagénomique, c'est-à-dire l'attribution des lectures à leur organisme d'origine dans un échantillon complexe. Cependant, deux obstacles empêchent le développement et la validation d'outils de binning exploitant cette information. D'une part, les signaux cinétiques bruts (deux valeurs uint8 par base et par strand) occupent un volume de stockage important et sont fréquemment supprimés lors du traitement et du partage des données, ce qui réduit le volume de données réelles disponibles pour entraîner et évaluer de tels outils. D'autre part, aucun simulateur de reads existant (PBSIM3 [4], BadRead [5]) ne reproduit ces signaux, rendant impossible tout *benchmarking* contrôlé.

Ce travail présente **KinSim**, un outil de génération de signaux IPD et PW biologiquement réalistes pour des reads PacBio HiFi simulés. KinSim repose sur un *conditional GAN* (cGAN) entraîné selon la formulation Wasserstein avec pénalité de gradient (WGAN-GP) [6]. Le générateur est un Transformer à blocs AdaLN-Zero [7] (~10 M paramètres) qui prend en entrée une fenêtre de séquence locale de 21 bases ainsi que les annotations de méthylation par position, et produit conjointement les quatre canaux cinétiques (IPD et PW pour les deux brins). Le conditionnement sur la méthylation s'effectue via une modulation affine par feature (FiLM [8] / AdaLN-Zero), permettant au modèle d'adapter sa distribution de sortie au type de modification présent. Un critique Transformer (3 M paramètres) avec normalisation spectrale apprend en parallèle à distinguer les signaux générés des signaux réels. Le modèle est entraîné sur 65 souches bactériennes : 52 Streptomyces séquencés sur PacBio Revio, et 13 isolats d'un run PacBio Vega, tous en chimie P1-C1.

Les signaux générés sont injectés dans des reads simulés par PBSIM3, produisant des fichiers BAM portant les tags cinétiques standards PacBio (fi/fp/ri/rp), directement compatibles avec les outils d'analyse en aval (pbmm2, ipdSummary, pbmotifmaker, Jasmine, modkit). KinSim permet ainsi de mettre en place des *benchmarks* contrôlés des outils de binning métagénomique exploitant l'information de méthylation, avec une vérité terrain connue pour chaque read.

Mots clés: Machine Learning, séquençage PacBio HiFi, méthylation bactérienne, signaux cinétiques, simulation de reads, binning métagénomique

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Abstract

KinSim: Development of a long-read simulation tool integrating kinetic signals

PacBio HiFi sequencing produces, alongside the DNA sequence, two per-base kinetic signals — the Inter-Pulse Duration (IPD) and the Pulse Width (PW) — whose variations reflect chemical modifications of the sequenced template, in particular bacterial methylations m6A, m4C and m5C [1], [2]. Because each bacterial species carries a distinct repertoire of methyltransferases [3], these marks constitute a species-specific signature that could improve metagenomic binning — the process of assigning reads to their organism of origin in complex samples. However, two obstacles prevent the development and validation of methylation-aware binning tools. First, raw kinetic signals (two uint8 values per base per strand) require substantial storage and are routinely discarded during data processing and sharing, drastically reducing the volume of real data available for training and validating such tools. Second, no existing read simulator (PBSIM3 [4], BadRead [5]) reproduces these signals, making controlled benchmarking impossible.

This work presents KinSim, a tool that generates biologically realistic IPD and PW signals for simulated PacBio HiFi reads. KinSim is built around a conditional Generative Adversarial Network (cGAN) trained under the Wasserstein-with-gradient-penalty (WGAN-GP) formulation [6]. The generator is a Transformer with AdaLN-Zero blocks [7] (~ 10 M parameters) that takes as input a 21-base local sequence window together with per-position methylation annotations, and produces the four kinetic channels (IPD and PW on both strands) jointly. Conditioning on methylation is achieved through feature-wise affine modulation (FiLM [8] / AdaLN-Zero), allowing the model to adapt its output distribution to the modification type present in the context. A Transformer critic (3 M parameters) with spectral normalisation learns in parallel to distinguish generated from real signals. The model is trained on 65 bacterial strains: 52 Streptomyces sequenced on PacBio Revio, and 13 isolates from a single PacBio Vega run, all in P1-C1 chemistry.

Generated signals are injected into reads simulated by PBSIM3, producing BAM files carrying the standard PacBio kinetic tags (fi/fp/ri/rp), directly compatible with downstream analysis tools (pbmm2, ipdSummary, pbmotifmaker, Jasmine, modkit). KinSim thereby enables controlled benchmarking of methylation-aware metagenomic binning, with known ground truth for every read.

Keywords: Machine Learning, PacBio HiFi sequencing, bacterial methylation, kinetic signals, read simulation, metagenomic binning

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1 Introduction

In bacteria, DNA methylation is not merely an epigenetic mark, it is the backbone of a defence system. Restriction-modification (R-M) systems consist of two enzymes that recognise the same short sequence motif: a methyltransferase that methylates the motif on the host DNA, protecting it, and a restriction endonuclease that cleaves any occurrence of the motif that lacks this methylation, effectively destroying unprotected foreign DNA such as bacteriophage genomes [1]. The most abundant class, Type II R-M systems, typically recognises palindromic motifs of 4–8 base pairs, sequences that read identically on both strands, and methylates them symmetrically [2]. Because each bacterial species carries a distinct repertoire of methyltransferases, each genome exhibits a unique methylation fingerprint: a specific combination of motifs, modification types (m6A, m4C, m5C), and stoichiometric occupancies [3]

PacBio Single Molecule, Real-Time (SMRT) sequencing captures this fingerprint as a byproduct of the sequencing process itself. As the polymerase synthesises DNA in a zero-mode waveguide, it produces two kinetic signals at each incorporation event: the Inter-Pulse Duration (IPD) and the Pulse Width (PW). When the polymerase encounters a methylated base on the template strand, its kinetics are altered, typically slowed, producing a measurable deviation from the expected signal [9]. The magnitude of this deviation depend on the modification type, [10] demonstrated that approximately 80% of IPD variance in unmodified DNA is explained by the local sequence context in a $[-7, +2]$ window around the incorporation site, establishing that kinetic signals are highly structured and predictable.

This species-specific methylation information, embedded directly in sequencing reads, opens a promising avenue for metagenomic analysis. In metagenomics, sequencing reads from complex environmental or clinical samples must be assigned to their organism of origin, a process known as binning. Current binning tools, notably, rely on sequence composition (tetranucleotide frequency) and differential coverage across samples, but these signals are often insufficient to resolve closely related species. Methylation patterns offer an orthogonal, organism-specific signal that could significantly improve binning resolution. A correction tool developed at the University of Fribourg aims to demonstrate this principle by using methylation density profiles to refine metagenomic bins [Navarro-Pau, unpublished]. However, validating such tools requires benchmarking on datasets where the true species origin of every read is known and where realistic kinetic signals are present.

Here lies the fundamental obstacle. Standard PacBio read simulators: PBSIM3 [4], BadRead [5], and others, generate reads with realistic sequence characteristics (error profiles, length distributions, quality scores) but produce no kinetic information whatsoever. The simulated reads lack the IPD and PW tags that real instruments produce. Without synthetic kinetic signals, it is impossible to evaluate whether a methylation-aware binning tool correctly exploits epigenetic information, because there is simply no epigenetic information in the simulated data.

KinSim closes that gap by injecting biologically realistic IPD and PW values into PBSIM3-simulated reads using a **conditional generative model** trained on real PacBio Revio / Vega data. Rather than predicting a single per-position summary statistic, the model jointly samples the full 21-base × 4-channel kinetic window (fi, fp, ri, rp) conditioned on the local sequence context and a per-position methylation-state vector. The architecture is a **DiT-style transformer generator** trained as a conditional WGAN-GP, so molecular-scale stochasticity is captured non-parametrically through a latent noise input z rather than through an assumed Gaussian. Outputs are written with the canonical PacBio raw-HiFi tag layout (fi/fp/ri/rp), making the generated reads drop-in compatible with ccs-kinetics-bystrandify → pbmm2 → ipdSummary / jasmine, and therefore directly usable for end-to-end benchmarking of methylation-aware metagenomic binning with known ground truth

2 Context

2.1 Company Overview

The University of Fribourg (UNIFR) is a bilingual public research university located in the city of Fribourg, Switzerland. Founded in 1889, it is the only fully bilingual university in Switzerland, offering courses in both French and German. The university comprises five faculties and hosts approximately 10,000 students and 800 researchers across a broad range of disciplines.

Figure 1: Switzerland map [11]

This internship takes place within the Department of Biology, which is part of the Faculty of Science and Medicine. The Department of Biology conducts research spanning molecular biology, ecology, evolution, and biochemistry. The Bioinformatics Core Facility, led by Dr. Laurent Falquet, provides computational expertise and support to research groups within and beyond the department. The facility specializes in sequence analysis, genome assembly, metagenomics, and the development of bioinformatics pipelines. It is within this facility that the present project is conducted, combining bioinformatics, machine learning, and sequencing technology to address challenges in metagenomic analysis.

2.2 Project Context

2.2.1 Methylation

DNA is not merely a sequence of nucleotides; it carries an additional layer of information through chemical modifications that regulate gene expression and cellular function. Among these modifications, DNA methylation is one of the most widespread and biologically significant. Methylation involves the addition of a methyl group (–CH_3) to a specific nucleotide base, typically catalyzed by enzymes known as DNA methyltransferases (MTases). These enzymes recognize specific short sequence patterns (referred here after as motifs) within the genome and modify one of more base within that motif. This modification does not alter the underlying DNA sequence but changes its biochemical properties and its interactions with other proteins.

Figure 2: Representation of the main methylation types [12]

Three major types of DNA methylation are found in nature: N^6-methyladenine (6mA), N^4-methylcytosine (4mC), and 5-methylcytosine (5mC). In eukaryotes, 5mC is the dominant form and plays central roles in epigenetic modifications and genomic imprinting. Here after, we will be mainly referring to bacterial DNA. DNA methylation in bacteria can be used to understand its pathogenicity [13], "*genome defense, chromosome replication and segregation, nucleoid organization, cell cycle control, DNA repair and regulation of transcription (…) The formation of epigenetic lineages enables the adaptation of bacterial populations to harsh or changing environments and modulates the interaction of pathogens with their eukaryotic hosts*" [3]

In bacterial genomes, DNA methylation operates within the framework of restriction-modification (R-M) systems. These systems consist of two enzymes that recognise the same short sequence motif: a methyltransferase (MTase) that methylates the motif on the host DNA, protecting it, and a restriction endonuclease (REase) that cleaves any occurrence of the motif that lacks this methylation [1]. The net effect is that the bacterium's own DNA, which is methylated, is protected, while foreign DNA entering the cell, such as bacteriophage genomes or unmodified plasmids, is recognised as unprotected and destroyed. Because bacterial DNA is not packaged into nucleosomes or compacted by histones, unlike eukaryotic DNA, it is directly accessible to both MTases and REases, resulting in methylation patterns that are more uniform and more directly tied to the activity of specific enzymatic systems than in eukaryotes.

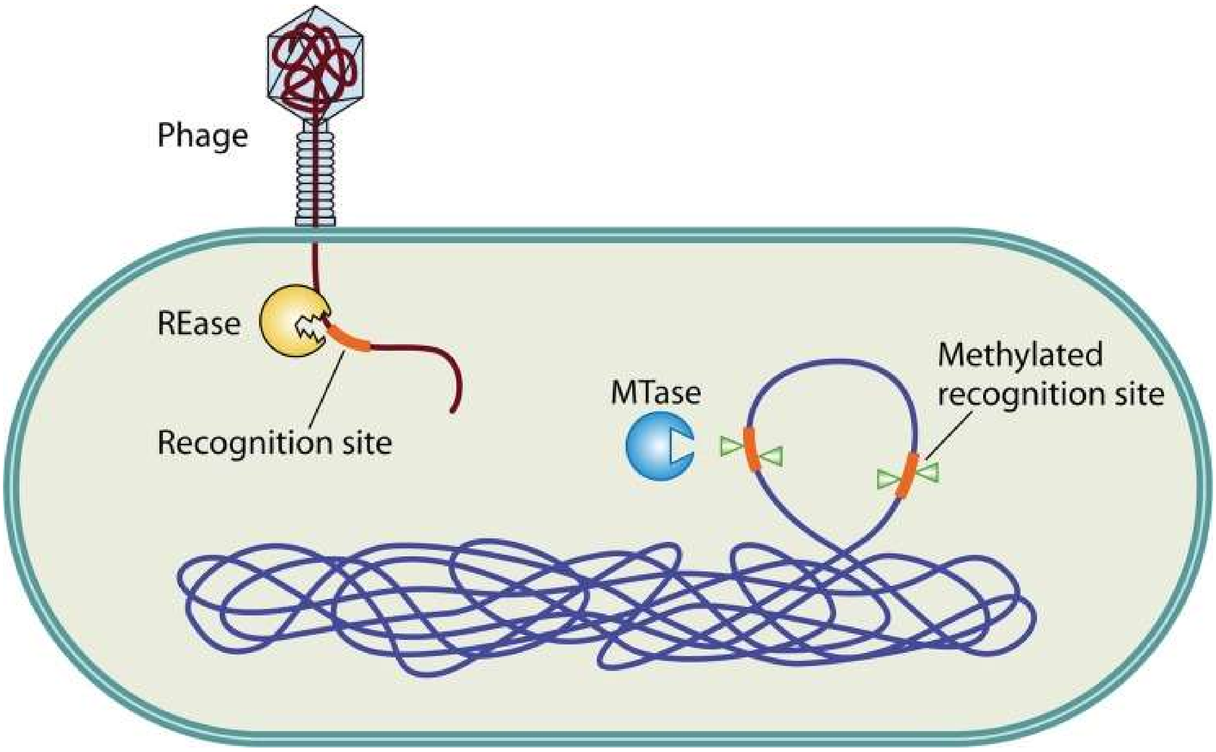


Figure 3: R-M system diagram [1]

R-M systems are classified into four types based on their subunit organization, cofactor requirements, and cleavage behavior [2]. Type I systems are the most complex, consisting of three subunits (R, M, S) that form a single multifunctional complex. They recognise bipartite asymmetric motifs (e.g. $AACN_6GTGC$) and cleave DNA at variable distances, often over 1,000 bp from the recognition site. Type II systems are the most prevalent, found in approximately 39% of sequenced bacterial genomes [1]. They consist of two separate enzymatic activities, an REase and an MTase, that act independently. Their target motifs are short sequences of 4–8 bp that are typically palindromic: the same sequence is read on both strands in the 5'→3' direction, and the MTase methylates the corresponding base on each strand symmetrically. This palindromic symmetry arises because the protein subunits form homodimers that recognise the same motif on both strands simultaneously. Type II systems are the dominant class in the data used in this project. Type III systems recognise asymmetric motifs, with the REase and MTase forming a single complex that cleaves DNA 25–27 bp from the recognition site. Type IV systems are functionally inverted: rather than protecting methylated DNA, they specifically cleave methylated sequences, serving a distinct biological role.

The motifs recognised by R-M systems vary between species and even between strains. Each bacterium carries a specific repertoire of MTases, each targeting a distinct motif and producing a specific type of modification (m6A, m4C, or m5C). This combination of motifs, modification types, and methylation fractions constitutes a species-specific methylation fingerprint [3], the biological foundation on which methylation-aware metagenomic binning relies.

The methylation state of a genomic position is not always binary. Stoichiometric variation arises because not every occurrence of a motif in every cell of a population is necessarily methylated at a given time. Phase variation, the reversible switching of gene expression, can lead to subpopulations with different methylation states at specific loci. Competition between MTases and REases for overlapping motifs, DNA replication timing (newly synthesised strands are transiently unmethylated), and growth conditions all contribute to fractional methylation. When sequenced, a motif site may appear fully methylated, fully unmethylated, or partially methylated across the population of reads covering that position.

2.2.2 PacBio SMRT

Pacific Biosciences (PacBio) Single Molecule, Real-Time (SMRT) sequencing is a third-generation sequencing technology that observes the activity of a single DNA polymerase molecule in real time. Unlike some short-read technologies that sequence amplified DNA fragments in bulk, SMRT sequencing captures the incorporation of each nucleotide as happens, producing long reads with rich kinetic information.

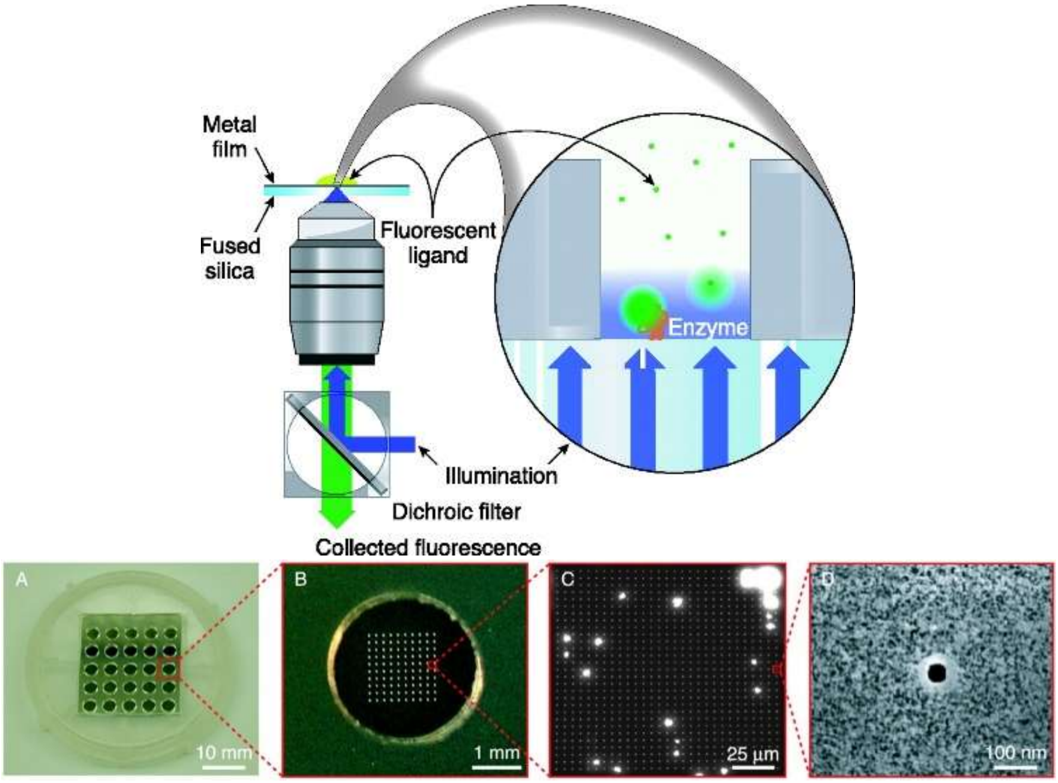


Figure 4: Zeromode waveguide representation [14]

PacBio SMRT produces High Fidelity reads. The polymerase is fixed in a Zero-mode waveguide. The core of the technology is the zero-mode waveguide (ZMW). A ZMW is a nanoscale hole, approximately 70 nm in diameter (Figure 4), fabricated in a thin metal film deposited on a glass substrate. The diameter of the ZMW is smaller than the wavelength of the excitation light, which means that light cannot propagate through the structure. Instead, an evanescent wave illuminates only the bottom few tens of nanometers of the well. A single DNA polymerase molecule is immobilized at the bottom of each ZMW, and the sequencing reaction takes place within this illuminated volume. Fluorescently labelled nucleotides diffuse into the ZMW, when a nucleotide is incorporated by the polymerase, it is held in the detection volume long enough to emit a characteristic fluorescent pulse. The color of the pulse identifies the base, while the duration and timing of the pulse encode kinetic information about the incorporation event.

The DNA template is prepared as a SMRTbell library (Figure 5): a linear double-stranded DNA molecule with single-stranded hairpin adapters ligated to both ends, forming a topologically circular structure. The polymerase processes this circular template continuously. In Continuous Long Read (CLR) mode, the polymerase makes a single pass around the template, producing reads that can exceed 100 kb but with a relatively high error rate (approximately 12-15%).

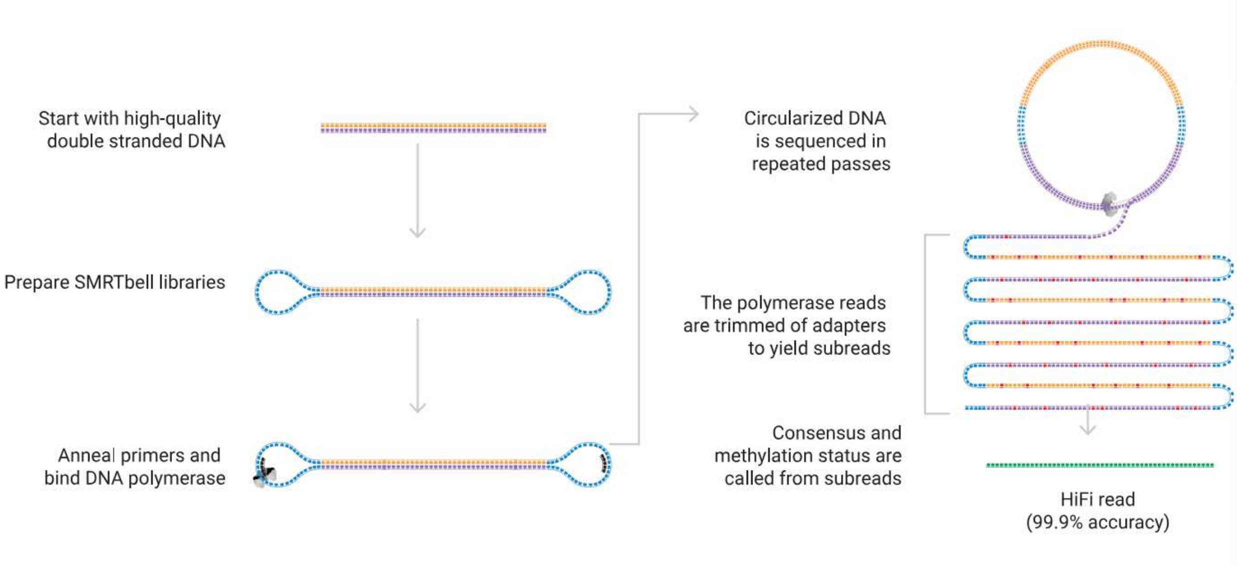


Figure 5: SMRT Bell [15]

In Circular Consensus Sequencing (CCS), which produces HiFi reads, the polymerase makes multiple passes around the same SMRTbell template (typically 3 to 30 passes depending on insert length). The resulting subreads are computationally combined into a consensus sequence with very high accuracy (>Q30, i.e., >99.9% accuracy). CCS/HiFi reads are typically 10-25 kb in length and represent the current standard for PacBio metagenomic sequencing.

What is the link with methylations?

During sequencing, PacBio instruments record two kinetic measurements at every nucleotide incorporation event (Figure 6): the Inter-Pulse Duration (IPD), which is the time elapsed between two consecutive fluorescent pulses, and the Pulse Width (PW), which is the duration of the pulse itself.

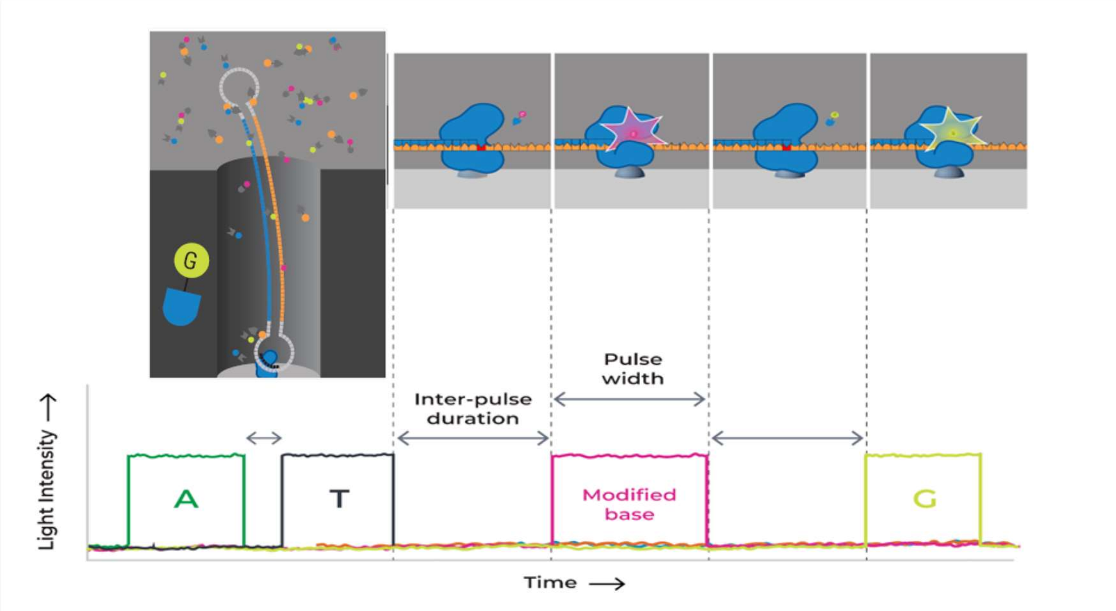


Figure 6: nucleotide incorporation kinetics in real time [16]

The speed of the polymerase varies based on the methylation state of the DNA, thus, IPD and PW tags vary significantly when the base being sequenced is methylated and has a visible variation based on the kmer context around. Seeing that variation at multiple similar sites, it is possible to find patterns (motifs) that the methyltransferases enzyme recognizes and methylates.

When the polymerase encounters a methylated base on the template strand, the methyl group alters the geometry of the base pairing interface, causing the polymerase to pause or slow down at characteristic positions relative to the modification. This perturbation produces a measurable deviation in both IPD and PW compared to the expected signal for unmodified DNA. The magnitude and spatial extent of this deviation depend on the modification type (Figure 7). For m6A, the main kinetic effect recognizable by PacBio technology is strongest at position 0 (the modified base itself) and at position +5 downstream. For m4C, the main effect is concentrated at position 0 only. For m5C, the main effects appears at positions +2 and +6 downstream, with additional subtle perturbations extending upstream to positions −2 to −10 [10].

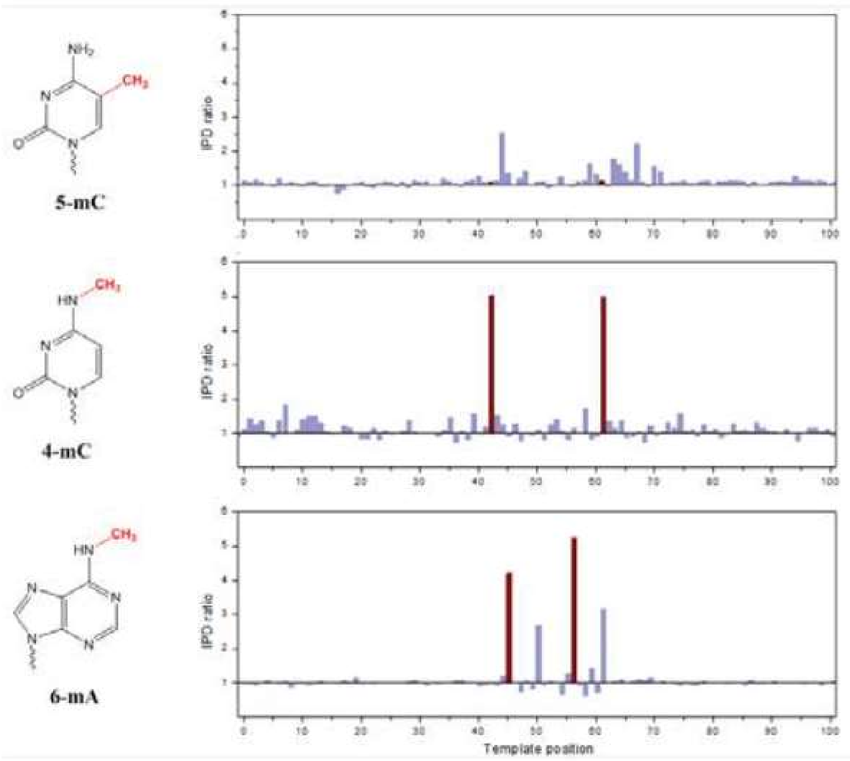


Figure 7: Influence of methylation type [9]

Raw IPD values, measured in frames, follow an approximately exponential distribution [9]. PacBio's encoding codec applies a non-linear compression that maps these frame counts to an 8-bit unsigned integer range (0–255), where 255 represents a cap on the maximum recordable value. This compression produces a roughly log-transformed representation in which per-k-mer distributions become approximately Gaussian. The overall IPD distribution can vary significantly between sequencing runs (referred to as movies), even for identical DNA samples, due to differences in reagent lots, temperature, instrument calibration, and SMRTcell variability.

The output of PacBio sequencing is stored in the BAM (Binary Alignment/Map) format, following the PacBio-specific BAM specification. Each read record contains the nucleotide sequence, quality scores, and kinetic tags. The fi:B:C (forward IPD) and fp:B:C (forward pulse width) tags store the per-base kinetic values as unsigned 8-bit integers (0–255, codec-compressed frame counts). Complementary-strand kinetics are stored in the ri:B:C and rp:B:C tags when available. On HiFi reads, these values represent averages across all subreads from the CCS consensus process. This averaging improves the signal-to-noise ratio for basecalling but attenuates the single-molecule kinetic deviations caused by methylation, making modification detection from HiFi reads more challenging than from raw subreads

2.2.3 PacBio Chemistry

Figure 8: Non-exhaustive PacBio chemistry generation timeline

The sequencing chemistry has evolved across PacBio instrument generations and directly affects kinetic measurements. Each chemistry version involves a specific polymerase enzyme and nucleotide formulation, identified by codes, the most relevant ones for the project are depicted in Figure 8. Different polymerase variants have different baseline kinetics and different sensitivities to methylation. The Revio platform uses the most recent chemistry, which is optimised for HiFi sequencing throughput. Those different chemistries are important because of the compatibility, or not, of specific tools and to know their limitations.

2.2.4 Methylation Detection Tools

Several computational tools exist for detecting methylation from PacBio kinetic data. ipdSummary [17] detects m6A and m4C from aligned subreads. pbmotifmaker [17] aggregates these calls into consensus motifs. Jasmine[18] detects 5mC and m6A directly from HiFi reads. modkit [19] aggregates per-read calls into site-level statistics. These tools are described in detail in Section 4.3.

2.2.5 Binning

Metagenomics is the study of genetic material recovered directly from environmental or clinical samples, without isolating individual organisms. A metagenomic sample contains DNA from potentially hundreds of different species mixed. After sequencing, the reads are assembled into contiguous sequences (contigs) using metagenomic assemblers such as hifiasm or metaFlye, which are designed to handle the complexity of mixed-species data.

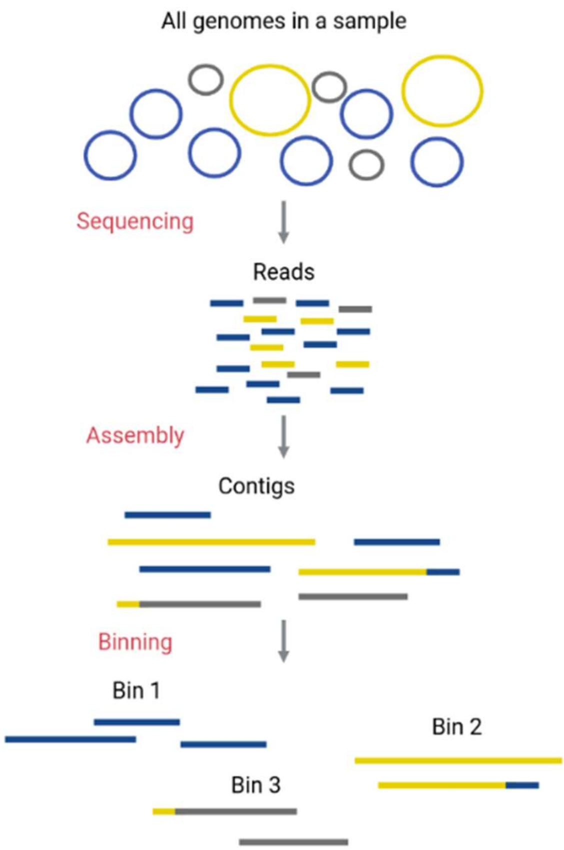


Figure 9: Binning logic [20]

The resulting contigs must then be grouped or binned by their organism of origin. Binning algorithms attempt to cluster contigs that belong to the same genome using several types of information. Sequence composition features, such as tetranucleotide frequency (TNF), capture the characteristic base composition of each organism's genome. Differential coverage across multiple samples exploits the fact that contigs from the same organism tend to co-vary in abundance. Tools such as MetaBAT2, MaxBin2, and SemiBin2 implement different algorithmic approaches (density-based clustering, expectation-maximisation, deep learning) to partition contigs into bins representing putative genomes. Consensus approaches like DAS Tool combine the outputs of multiple binning algorithms to improve accuracy.

Despite significant progress, metagenomic binning remains imperfect. Closely related species with similar sequence composition are difficult to separate. Low-abundance organisms produce fragmented assemblies with short contigs that lack sufficient signal for reliable binning. Horizontal gene transfer introduces genomic regions with atypical composition that may be misassigned. The accuracy of binning directly affects all downstream analyses, including taxonomic profiling, functional annotation, and the reconstruction of metagenome-assembled genomes (MAGs). Any additional source of organism-specific information that can complement sequence composition and coverage has the potential to improve binning quality.

2.2.6 Correction tool

A correction tool was developed as a prior project at the University of Fribourg (Correction-Tool, by A. Navarro-Pau) to improve metagenomic binning by incorporating methylation information. The tool operates in several steps: it first runs PacBio's SMRT Link analysis pipeline (ipdSummary and pbmotifmaker) to detect methylation motifs in the metagenomic sample. It then counts, for each contig, the total number of motif occurrences and the number of methylated occurrences using EMBOSS fuzznuc. The ratio of methylated to total motif sites per contig produces a methylation density profile. Finally, contigs are clustered based on these methylation profiles using UMAP dimensionality reduction followed by DBSCAN clustering and the resulting clusters are used to refine the original bins.

The rationale is straightforward: contigs originating from the same organism share the same methyltransferase repertoire and therefore exhibit the same methylation motif patterns and densities. Two contigs from the same species will both show high methylation density at GATC sites (if that organism has a GATC-targeting m6A methyltransferase), while contigs from a different species may show no methylation at GATC but high methylation at CCWGG. This organism-specific methylation fingerprint provides an orthogonal signal to sequence composition and coverage.

However, the correction tool has not been fully validated because of a fundamental data limitation: there is no ground-truth dataset of metagenomic PacBio reads with known methylation patterns. Real metagenomic samples have unknown composition, and mock communities do not preserve kinetic information in the distributed data. PacBio instruments store kinetic tags (IPD and PW) during sequencing, but these tags are large (two uint8 values per base per read) and are routinely discarded during data processing and sharing to reduce storage costs. Furthermore, PacBio and other sequencing providers do not plan to systematically preserve kinetic data in public repositories.

This creates a need for a simulation tool capable of generating realistic kinetic signals for synthetic metagenomic reads. Given a set of simulated reads (produced by tools such as PBSIM3) and knowledge of the methylation motifs present in each organism, such a tool would inject biologically plausible IPD and PW values into the reads, producing BAM files indistinguishable, from the binning tool's perspective, from real sequencing data. This is the purpose of KinSim.

2.3 Specifications / Requirements

Existing PacBio HiFi read simulators, PBSIM3, BadRead, (…) faithfully reproduce sequence-level characteristics such as error profiles, read length distributions, and coverage patterns. However, none of them simulate per-base polymerase kinetics. Simulated reads lack the IPD and PW tags that real PacBio instruments produce, making it impossible to benchmark any tool that relies on kinetic signals. Furthermore, raw kinetic data from real sequencing runs is frequently discarded during processing and sharing due to its storage cost, limiting the availability of real-world data for developing and validating such tools.

KinSim addresses this gap. It takes reads already simulated by PBSIM3 and injects biologically realistic IPD and PW values using a neural network trained on real PacBio Revio and Vega data. The result is a simulated BAM file that is format-compatible with real sequencing output, but with known ground truth for both sequence origin and methylation state.

- **Inputs**

KinSim requires two inputs. The first is a reference genome in FASTA format, representing the organism to be simulated. The second is a set of methylation motifs: short sequence patterns (e.g. GATC for m6A, CCWGG for m5C) along with the type of modification and the position of the modified base within the motif. These motifs can be obtained from the REBASE database [21], extracted from real data using the upstream calling pipeline (ipdSummary, Jasmine, modkit), or defined manually.

- **Outputs**

KinSim produces unaligned BAM files containing PacBio HiFi reads with embedded kinetic tags. Each read carries fi:B:C (forward IPD) and fp:B:C (forward pulse width) tags encoded as unsigned 8-bit integers following the PacBio BAM specification. These files are directly compatible with downstream analysis tools such as pbmm2, ipdSummary, Jasmine, and modkit.

- **Scope**

The current implementation targets PacBio HiFi data. All training data originates from the Revio and Vega platforms using P1-C1 chemistry. KinSim supports three types of DNA methylation: m6A, m4C, and m5C. The methylation alphabet is configuration-driven: adding a new modification type requires only a YAML change, not a code modification. The tool is designed for prokaryotic genomes, where methylation motifs are well-defined and species-specific. Extension to eukaryotic methylation is architecturally possible but not validated.

- **What KinSim does not do**

KinSim is not a read simulator, it does not generate DNA sequences, simulate error profiles, or model read length distributions. These tasks are handled upstream by PBSIM3. KinSim is also not a methylation caller, it does not detect modifications from real sequencing data. Its role is strictly generative: given a sequence and a methylation configuration, it predicts what the kinetic signals would look like if that sequence were sequenced on a PacBio instrument.

3 State of the Art

The goal is to generate realistic numeric data distributions that copies the overall constraints of the polymerase kinetics when sequencing DNA that can be methylated.

3.1 Per-Key Statistical Models

The most direct approach to modelling kinetic signal distributions is to estimate, for each unique combination of sequence context and methylation state, a parametric or non-parametric distribution of the observed IPD and PW values. Each unique configuration forms a *key*, and a separate distribution is fitted from the samples sharing that key. ipdSummary, the canonical first-generation methylation caller for PacBio data, takes a **tree-based statistical approach**: a gradient-boosted regression-tree ensemble is trained per sequencing chemistry to predict the expected unmodified IPD from the local 12-mer context, and a per-position modification call is made by comparing the observed IPD to that prediction under a one-sided test.

Several statistical modelling techniques can be applied at this stage. Kernel Density Estimation (KDE) builds a continuous non-parametric distribution from observed samples without assuming a specific functional form, smoothing the empirical samples by a kernel function (typically Gaussian) to produce a continuous density. Gaussian Mixture Models (GMM) fit a finite combination of Gaussian components, allowing the distribution to capture multimodal behaviour, particularly relevant for PacBio kinetics, where polymerase activity exhibits multiple regimes (e.g., the slow and fast tau states that produce bimodal IPD distributions [22]. GMMs are more compact than KDE, they store only a small number of parameters per key, and the optimal number of components can be selected automatically via the Bayesian Information Criterion (BIC). Quantile Regression offers a third option, capturing the conditional distribution at specified quantiles. All three methods share the same underlying logic: one key, one independently fitted distribution.

The choice of *what* constitutes the key is not arbitrary and is the central methodological question of this category. A simple approach would key only on methylation type at the centre position (one distribution for unmodified, one for m6A, one for m5C, etc.) and emit values from this distribution regardless of the surrounding sequence. This is rejected for two compounding biophysical reasons. First, the kinetic response of the polymerase depends strongly on the local sequence context: the same m6A modification embedded in two different k-mers produces different IPD intensities, because the polymerase response reflects its interaction with the entire sequence neighbourhood, not the modified base in isolation. Second, methylation effects propagate beyond the position of the modified base itself: m6A produces its main IPDratio peaks at offsets 0 and +5, m5C at offsets +2 and +6, m4C at offset 0 [9]. The IPD value at any given position therefore depends on the methylation states of *all* positions in a local window around it, not just the central one. The appropriate key for kinetic modelling is thus the combination of the local sequence (k-mer) and the full methylation pattern in the local window, a structure that captures both the sequence-dependent baseline kinetics and the cumulative effect of methylations at biophysically relevant offsets.

This per-key formulation is conceptually sound and has proven effective for the *detection* of methylation in real sequencing data, where each observed position is compared against its key-specific empirical distribution. For the *generation* task addressed in this work, however, it suffers from a fundamental limitation: it cannot generalize beyond the keys present in the training data. The combinatorial space of possible (k-mer, methylation pattern) keys is astronomically large, with $4^{11} \approx 4.2$ million possible for 11-mers and exponentially many methylation patterns over a window of size 11, and only a vanishingly small fraction of these keys is observed in any practical dataset. A per-key model can only emit values for keys it has seen; for any unseen sequence-methylation combination, it has no prediction to offer. The model treats each key as an isolated unit and does not learn the underlying relationship between sequence composition, methylation pattern, and kinetic response. The kinetic signal observed at GATCGCATCAA cannot inform the prediction at GATCGCATCAG, even though the two contexts share ten of eleven positions and likely produce nearly identical kinetic responses. Recovering this latent relationship, and thereby enabling generalization across the full space, requires a model capable of *learning a representation* of the input rather than memorizing distributions per key. This is the role of machine learning, developed after this next section.

3.2 Sequential Methods

Hidden Markov Models (HMMs) are a natural candidate for sequential data generation: at each position, a latent state governs the emission distribution, and transition probabilities control the progression from one state to the next. Applied to kinetic signal prediction, the hidden states would represent the methylation status (unmethylated, m6A, m4C, m5C) and the emissions would be the IPD and PW values at each position. However, HMMs are fundamentally unsuited to this task for two reasons. First, the emission at each state is context-independent — the model draws from the same distribution regardless of the surrounding DNA sequence, discarding the dominant source of kinetic variance. Second, methylation is not a persistent state: the kinetic effect is confined to a few specific offsets, with hundreds or thousands of unmodified positions between consecutive methylated motifs, leaving no meaningful sequential dependency for the transition matrix to capture.

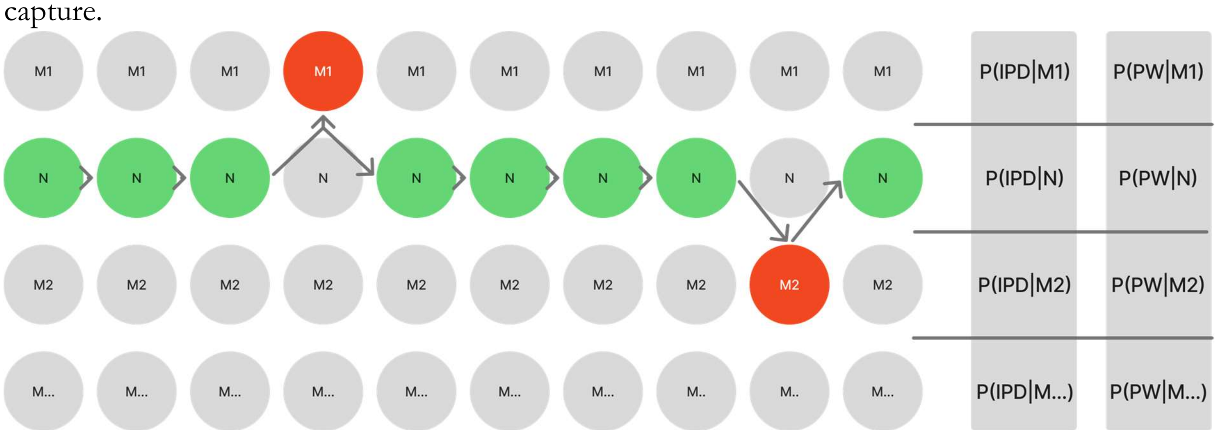


Figure 10: HMM simple representation for methylation

The figure above illustrates how an HMM models kinetic signal prediction. Each row represents a hidden state: M1 (methylation type 1, e.g. m6A), N (no methylation), M2 (methylation type 2, e.g. m4C), and so on. At each position along the DNA, the model transitions between states and emits an IPD and PW value drawn from a state-specific distribution. The green nodes indicate positions where a methylation state is active; the red node marks the single position where the kinetic effect manifests before the model transitions back to the normal state. This reveals the core problem: methylation main recognizable effects on HiFi reads are limited to specific bases (Not exhaustive), offsets 0 and +5 for m6A, offset 0 for m4C, offsets +2 and +6 for m5C, that are mostly non-adjacent. Between two methylated motif sites, hundreds or thousands of unmodified positions intervene. There is no sequential dependency to exploit: the model would spend nearly all its time in the normal state, with rare, isolated excursions into a methylation state that immediately returns to baseline. The transition matrix collapses to near-deterministic values, eliminating the sequential modelling that makes HMMs powerful. Each state emits from a single distribution regardless of surrounding sequence context, discarding the dominant source of signal variance.

Historically, HMMs did play a central role in PacBio's technology, not for methylation detection, but for basecalling. Early PacBio basecallers used HMMs to decode raw fluorescence pulses into DNA sequences, where the hidden states represented the four nucleotides and the emissions were fluorescence intensities. This is a natural application: consecutive bases form a genuine sequential chain, and the transition probabilities capture the dynamics of polymerase incorporation. These HMM-based basecallers have since been replaced by neural network approaches, first CRF/RNN architectures, then DeepConsensus [23] which achieve higher accuracy by learning richer representations of the signal. For methylation detection specifically, HMMs were never adopted for methylation detection: existing kinetic analysis tools occupy a narrow design space. ipdSummary performs per-position statistical testing against a context-conditioned expected IPD predicted by a gradient-boosted regression model over a 12-mer. Modern callers such as Jasmine replace that with a CNN-based per-position classifier. Both are discriminative tools — they decide whether a position is modified — and neither generates kinetic signals.

3.3 Machine Learning: Deep Learning Approaches

The limitations identified in Sections 3.1 and 3.2 converge on a common requirement: a model capable of learning a *representation* of the local sequence and methylation context, from which kinetic signals can be predicted in a generalizing way. Deep learning provides this capability.

Unlike per-key statistical models, which memorize one distribution per observed key, a neural network learns a continuous mapping from the input space (the k-mer sequence, the methylation pattern, the reverse-strand methylation) to the output space (the IPD and PW distributions at the active site). The model parameters are shared across all training samples, so information learned about one (k-mer, methylation pattern) configuration informs the prediction for all related configurations. A k-mer that shares ten of its eleven positions with a previously seen example is no longer an unseen key, it is a nearby point in a learned representation space, and the model can interpolate accordingly. This is the fundamental advantage of representation learning over per-key estimation.

The limitations of both per-key statistical models and HMMs converge on the same requirement: a model that learns a continuous representation of the input rather than memorising per-key distributions or relying on state transitions. The following section reviews the deep learning architectural choices available for this task.

3.3.1 Deep Learning and Genomics

- **Problems specification**

The task at the core of this project is to predict, for each base in a DNA read, the kinetic signals (IPD and PW) that a PacBio polymerase would produce. This prediction must account for two sources of variation: the local sequence context, which dominates the signal, explaining approximately 80% of IPD variance in unmodified DNA and the methylation state, which introduces additional kinetic deviations at and around modified bases.

The observed signal at a given position is not deterministic: for the same k-mer and methylation state, IPD and PW vary molecule-to-molecule because of stochastic polymerase kinetics and uint8-codec quantisation. A useful generative model must therefore sample from a distribution rather than emit a point estimate.

This formulation leads to four sub-problems that any deep learning architecture must address: **(1)** representing discrete DNA bases as numerical inputs the network can process, **(2)** extracting the local sequence patterns that drive polymerase kinetics, **(3)** conditioning the prediction on the methylation state, and **(4)** outputting signals that comes from the data distribution.

1. **Representing DNA**

Neural networks operate on continuous numerical vectors, not on discrete characters. The first design decision is therefore how to encode the four DNA bases (A, C, G, T) as input to a model.

The simplest approach is **one-hot encoding**, where each base is represented as a binary vector of length four (e.g. A = [1,0,0,0], C = [0,1,0,0]). This representation is unambiguous and makes no assumptions about relationships between bases, but it is sparse and uninformative: the model must learn from scratch that, for instance, purines (A, G) share biochemical properties. One-hot encoding is used in several genomic deep learning tools, including DeepSignal [24] and DeepMod [25]

A more expressive alternative is **learned embeddings**, where each base is mapped to a dense vector of arbitrary dimensionality through a trainable lookup table. During training, the embedding vectors are adjusted so that bases with similar functional roles end up with similar representations. This approach is standard in natural language processing and has been adopted in genomics by models such as Enformer [26] and DNABERT.

Positional embeddings: fixed or learned vectors added to the base embeddings, encode the relative location of each base within the input window. This is particularly relevant for polymerase kinetics, where bases upstream and downstream of the incorporation site have asymmetric effects: Feng et al. (2013) showed that the 7 bases upstream have a much stronger influence on IPD than the 2 bases downstream.

Embeddings can also be defined at the **k-mer level** rather than the single-base level. DNABERT tokenizes DNA into overlapping k-mers (k = 3–6), treating each k-mer as a single token. DNABERT-2 [27] replaces k-mer tokenization with Byte Pair Encoding (BPE), a data-driven approach that learns variable-length tokens from the training corpus. Those representations can after be used as a starting "general representation" of a kmer and be fine tuned to the output we desire.

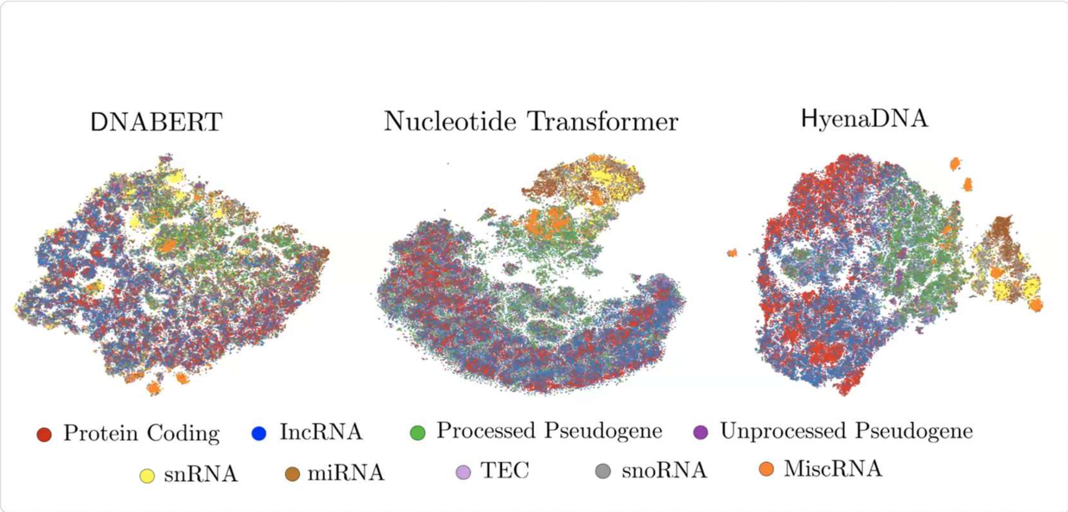


Figure 11: Visual Representation of categories in transformer representations

Those models, while heavy, encodes a high-dimensionl property of DNA.

2. Extracting local patterns

The second design decision is which architecture to use for extracting relevant patterns from the encoded sequence. Several families of architecture have been applied to genomic signal prediction tasks.

- **Multi-layer perceptrons**

A **multi-layer perceptron (MLP):** MLPs have no spatial inductive bias and cannot exploit adjacency between bases.

- **Convolutional neural networks**

One-dimensional convolutional neural networks (1D CNNs) apply learned filters that slide along the sequence, detecting local motifs such as dinucleotides or trinucleotides regardless of their position. This translation-invariant property is well suited to DNA, where the same motif can appear anywhere in a read. Stacking multiple convolutional layers increases the effective receptive field, allowing the network to capture longer-range dependencies.

Jasmine, the current PacBio tool for 5mC and 6mA calling on HiFi reads, uses a CNN architecture that jointly models IPD, PW, and local sequence context. DeepSignal combines Inception-style CNN modules with recurrent layers for nanopore methylation detection. These examples demonstrate that convolutional architectures effectively capture the local sequence patterns that influence polymerase and pore kinetics.

- **Recurrent neural networks**

Recurrent neural networks (RNNs) process the sequence one base at a time, maintaining a hidden state that carries information from previous positions. This sequential process mirrors how a polymerase physically reads along the DNA strand. However, vanilla RNNs suffer from vanishing gradients and struggle to retain information over long distances.

Gated variants address this limitation. **Long Short-Term Memory (LSTM)** networks use input, forget, and output gates to selectively retain or discard information. **Gated Recurrent Units (GRUs)** achieve similar performance with fewer parameters by merging the forget and input gates.

Bidirectional variants (BiLSTM, BiGRU) process the sequence in both directions simultaneously, capturing context from both upstream and downstream of each position. This is important because polymerase kinetics depend on flanking bases in both directions. DeepSimulator [28] uses a BiLSTM with Deep Canonical Time Warping to translate DNA sequences into simulated nanopore electrical signals, the most established deep learning approach for sequence-to-instrument-signal simulation. KinMethyl [29] uses a BiGRU to predict IPD and PW from a 21 bp window for unmodified PacBio DNA, with two independent linear heads for each signal type. DeepMod applies BiLSTMs for 5mC and 6mA detection from nanopore data within 21-event windows.

- **Transformers**

Transformer architectures replace recurrence with self-attention, allowing every position to attend every other position simultaneously. This removes the sequential bottleneck of RNNs and enables fully parallelized training but introduces quadratic memory scaling with sequence length.

seq2squiggle [30] uses a non-autoregressive feed-forward transformer to predict nanopore signals from DNA sequence end-to-end, with support for methylation modifications (5mC, 5hmC). Rockfish [31] employs a transformer encoder-decoder for read-level 5mC detection from combined nanopore signal and reference sequence. MuLan-Methyl [31]uses an ensemble of five fine-tuned BERT variants to predict 6mA, 4mC, and 5hmC from DNA sequence alone.

3. Conditioning on auxiliary information

In many genomic prediction tasks, the output depends not only on the DNA sequence but also on categorical metadata — such as the type of base modification, the sequencing chemistry, or the cell type. Several strategies exist for injecting this auxiliary information into the network.

- **Concatenation**

Appends the metadata (as a one-hot vector or learned embedding) to the sequence features before passing them through subsequent layers. This is the simplest approach: the network receives the combined information and must learn on its own how to use it. The disadvantage is that the auxiliary signal may be diluted in high-dimensional feature vectors.

- **Addition**

Sums a metadata embedding directly with the sequence features, element-wise. This implicitly assumes that the auxiliary information acts as a uniform additive shift across all feature dimensions — a strong assumption that may not hold when different modifications affect different aspects of the signal.

- **Multiplication (element-wise)**

Scales the sequence features by the metadata embedding. This allows the auxiliary input to amplify or suppress specific features but cannot shift the baseline — the opposite limitation of addition.

- **Feature-wise Linear Modulation (FiLM)**

Combines both operations: it generates two vectors, γ (scale) and β (shift), from the auxiliary input and applies an element-wise affine transformation: output = $\gamma \cdot$ features + β. This is strictly more expressive than either addition or multiplication alone, as each feature channel can be independently scaled and shifted. FiLM was originally developed for visual reasoning tasks and has since been applied in genomics: FiLM layers are also used to condition gene expression predictions on perturbation type and cell line identity [30]. Temporal FiLM [32] explicitly extends FiLM to sequential data and lists genomic signal prediction as an application domain.

4. Outputting from a distribution

For a given sequence context and modification state, the observed kinetic signal varies molecule-to-molecule because of stochastic polymerase kinetics, codec quantisation, and measurement noise. A generative model must therefore sample from a distribution rather than emit a point estimate — and, since the polymerase traverses every position of a window in the same pass, that distribution is **joint over the full kinetic tile**, not a product of independent per-position marginals. The way this joint distribution is represented determines the model's expressiveness, its training objective, and its inference cost.

- **Point prediction with injected noise**

Predict a single value per position under mean-squared-error loss and add fixed-variance Gaussian noise at generation time. Cheap, but the noise is homoscedastic by construction — too wide on stable contexts, too narrow on noisy ones — and the per-position factorisation destroys the within-tile correlations.

- **Gaussian with learned variance**

Predict (μ, σ) per output and train under Gaussian negative log-likelihood. Each context gets its own variance and the model is penalised both for inaccurate means and for miscalibrated spread. This is the formulation KinMethyl uses for its point estimates (μ only, no σ head), and it is standard in probabilistic forecasting. It remains parametric, unimodal, axis-aligned, and still per-position independent unless an explicit covariance head is added, which scales quadratically in window length and is rarely well-conditioned.

- **Variational autoencoders**

(Kingma & Welling, 2013) encode the target into a continuous latent space and decode through a probabilistic decoder, with a variational lower bound as the training objective. Likelihood is tractable; the reconstruction-vs-regularisation trade-off typically blurs the decoded outputs.

- **Diffusion models**

(Ho et al., 2020) iteratively denoise a Gaussian into a sample and produce very sharp distributions, at the cost of tens to hundreds of forward passes per sample, a non-trivial budget when the target corpus is at the scale of a full sequencing run.

- **Adversarial sampling (GANs)**

[33] Inverts the problem in a way that closely mirrors the methodology originally outlined by Dr. Falquet for this project: rather than committing to a parametric form, the model is asked to look at real PacBio reads and decipher what the kinetic signature of each sequence-and-methylation context actually looks like, and then to draw new samples from that deciphered representation. The framework realises this in two coupled networks. A generator G maps a latent noise vector z into a candidate sample. A critic D learns, from real reads, to separate genuine kinetics from those produced by G. The two are trained against one another until G can produce samples that D can no longer distinguish from real ones — at which point D has effectively encoded "what methylated kinetics look like", and G has been forced to sample from the same representation. There is no parametric distribution committed to in advance, no per-position factorisation, and no iterative denoising; at convergence, a single forward pass through G draws from the data distribution directly. The cost is training stability: the original Jensen-Shannon objective is prone to mode collapse and vanishing gradients, both of which the Wasserstein formulation with gradient penalty [34] addresses by replacing the divergence with a 1-Lipschitz Wasserstein critic and penalising the critic's gradient norm.

The four criteria that matter for the present problem, a non-parametric joint over a short kinetic window, no commitment to a Gaussian shape on log-quantised data, intrinsic capture of within-window correlations, and one forward pass at inference time, single out the adversarial family.

3.3.2 Summary

The deep-learning toolkit for sequence-to-signal prediction in genomics has converged on a small set of well-understood building blocks. CNNs and BiGRU/BiLSTMs are effective at extracting local patterns from short DNA contexts; transformers extend that reach to longer-range dependencies and to joint structured output over a window. FiLM provides a principled way to inject categorical metadata such as modification type by scaling and shifting the sequence features per channel, rather than diluting the signal through plain concatenation. For the output side, the choice ranges from cheap point predictors with injected noise, through parametric likelihood heads (Gaussian-NLL mixtures), to implicit generative models, VAEs, diffusion, and adversarial sampling, that drop the parametric assumption entirely. Within that space, the requirements of the present problem, a non-parametric joint distribution over a short kinetic window, no commitment to a Gaussian shape on log-quantized data, intrinsic capture of within-window correlations, and a single forward pass at inference time, pull decisively toward the implicit-sampling end of the spectrum.

Despite the maturity of these building blocks, a gap remains in existing literature. DeepSimulator [28] and seq2squiggle [30] simulate nanopore squiggles but not PacBio kinetics. KinMethyl is the closest published model: a BiGRU that predicts IPD and PW from a 21 bp window on PacBio data. It is, however, trained exclusively on Whole-Genome Amplified (WGA) DNA, material where all methylation has been chemically erased, making it an unmodified-only baseline predictor. It does not condition on methylation type, does not produce a distributional output, and presupposes the availability of WGA training data, which is not generally the case. No existing model jointly predicts IPD and PW distributions conditioned on both sequence context and methylation type for PacBio HiFi data, nor does so without requiring WGA training data.

3.4 Tools and Pipeline

Training a kinetic signal predictor requires two categories of input: real PacBio data with methylation annotations (to learn from), and simulated reads without kinetic signals (to inject predictions into). This section describes the tools involved in preparing both.

3.4.1 Reference Genome Assembly

Some organisms used in this project did not have reference genomes available. In those cases, de novo assembly was performed using hifiasm [35], a haplotype-resolved assembler designed for PacBio HiFi reads. Hifiasm exploits the high accuracy of HiFi data (>Q30) to produce highly contiguous assemblies without the need for error correction, typically yielding contig N50 values in the megabase range for bacterial genomes. The resulting assemblies served as the reference input for all downstream steps: alignment, methylation calling, and KinSim training and generation.

3.4.2 Methylation calling / motifs from real PacBio BAMs

The quality of the methylation annotations directly determines the quality of the training data: mislabelled positions introduce noise that the model cannot distinguish from genuine signal. The calling pipeline combines four tools, each covering a different aspect of the detection problem.

IpdSummary, part of PacBio's kineticsTools package [17], detects m6A and m4C modifications from aligned subreads. For each genomic position, it compares the observed IPD against the expected IPD for unmodified DNA. This expected value is predicted by a gradient boosted regression tree model (GBM) trained on the 12-mer sequence context surrounding the active site. Several chemistry-specific GBM models are available, each trained on whole-genome amplified data from the corresponding chemistry: P6-C4 (RS II), SP2-C2 (Sequel), and SP3-C3 (Sequel). No model exists for the Revio P1-C1 chemistry used in this project. On the recommendation of PacBio scientists, the SP3-C3 model was selected as the closest available approximation to Revio kinetics. Positions where the observed IPD deviates significantly from the model prediction are assigned a Modification QV score and tagged with the most likely modification type. The output GFF file reports, for each called position, the modification type, the QV score, and the ±20 bp sequence context surrounding the site. The chemistry mismatch between the SP3-C3 model and the actual Revio data means that baseline IPD predictions are not perfectly calibrated, potentially reducing detection sensitivity, particularly for weak signals such as m5C. This limitation propagates into the KinSim training data: any methylation site missed or miscalled due to the chemistry mismatch introduces noise into the training annotations.

pbmotifmaker, part of the SMRT Analysis suite, takes the per-position calls from ipdSummary and identifies consensus methylation motifs using a greedy branch-and-bound search. For each motif, it reports the sequence in IUPAC notation, the modified position, the modification type, and the fraction of genomic occurrences that are detectably methylated.

1. Jasmine (Formerly primrose)

Figure 12: Jasmine Logo [18]

Jasmine (formerly Primrose) is PacBio's tool for detecting 5mC at CpG sites and m6A from HiFi reads. It is based on the HK model structure developed by Tse et al. [19].

The HK model (named after its authors Tse, Ho, and Korlach) takes a holistic approach: for each CpG site, it examines a measurement window of 21 nucleotides centred on the target cytosine. Within this window, it uses the IPD, PW, and base identity at every position, not just at the CpG itself. This design is critical because the kinetic effect of 5mC is weak and distributed across multiple neighbouring positions. Feng et al. [17] showed that polymerase kinetics at a given position are primarily influenced by the upstream 7 bases and downstream 2 bases, with additional perturbations extending to positions −10 and −2 relative to the modified base. These distributed effects motivate the use of a 21 bp context window for methylation calling.

For each targeted CpG, a 21-mer surrounding the base is extracted. The mean and standard deviation of IPD and PW values at each position in the 21-mer are calculated, forming a 5×21 feature matrix (sequence + mean IPD + std IPD + mean PW + std PW). Because CpG methylation is symmetric between strands, the feature matrices from both DNA strands are combined for prediction. A convolutional neural network is then trained on this feature matrix, using amplified DNA (unmethylated) and M.SssI-treated DNA (methylated) as training data. The area under the curve for differentiating methylation states reached up to 0.97.

A key limitation is that these methods rely on 21 bp contexts surrounding the target CpGs and have over 29% low-confidence (<75% accuracy) calls at CpGs with less distinguishable signals. Ni et al. (2024) hypothesised that incorporating CpG methylation correlation information at the single-molecule level could improve calls on these low-confidence CpGs, and presented hifimeth, a deep graph convolutional network that uses 400 bp context, achieving 94.7% accuracy — a 5.5% improvement over ccsmeth.

A fundamental limitation for this project is that Jasmine's 5mC detection is restricted to CpG dinucleotide contexts, a pattern prevalent in eukaryotic genomes but not representative of bacterial m5C, which occurs at diverse motifs such as CCWGG (Dcm methyltransferase). No per-read m5C caller currently exists for non-CpG contexts on PacBio HiFi data. Jasmine nonetheless remains the only available option for per-read 5mC calling and is used in this project with the understanding that its coverage of bacterial m5C motifs is incomplete.

2. Modkit

Figure 13: Nanopore logo [19]

Modkit is the de facto post-processing tool for the modified-base SAM tags MM and ML defined by the hts-specs standard, the per-read, per-base modification-probability format that jasmine emits for 5mC on PacBio reads. Originally developed by Oxford Nanopore for nanopore data, modkit operates strictly on the tag content and is therefore platform-agnostic: any aligned BAM carrying MM/ML tags can be processed, regardless of the underlying chemistry. Two sub-commands are used in this pipeline. modkit pileup aggregates the per-read probabilities at every reference position into a **bedMethyl** record, chromosome, start/end, modification code, coverage, methylated fraction, and per-strand counts, collapsing thousands of probabilistic per-molecule calls into a single consensus methylation fraction per genomic site. modkit find-motifs then takes that pileup and searches the local sequence context around methylated positions for enriched k-mer patterns, yielding the IUPAC motifs (e.g. CCWGG, GATC) under which methylation preferentially occurs in the sample. In the KinSim pipeline, modkit therefore plays the same role for jasmine's 5mC calls that pbmotifmaker plays for ipdSummary's m6A/m4C calls: it converts raw per-position evidence into the sample-level motif catalogue that downstream extraction and benchmarking consume.

3.4.3 Read Simulation

PBSIM3 is the de facto open-source simulator for long-read sequencing, supporting both PacBio (RS II / Sequel / HiFi-via-CCS) and Oxford Nanopore platforms. It reproduces the multi-pass nature of PacBio by emitting Continuous Long Reads (CLR), multiple subreads from repeated polymerase passes over the same circular template, which can then be fed through PacBio's CCS software to obtain HiFi consensus reads, mirroring the real data-generation workflow. The simulator covers the structural and error properties of the reads: sequence content, read-length distribution, subread count per molecule, and per-position substitution / insertion / deletion profiles, with quality profiles calibrated by default on Sequel chemistry.

What PBSIM3 does *not* simulate is the kinetic side-channel, the per-base Inter-Pulse Duration (IPD) and Pulse Width (PW) values that encode polymerase dynamics during synthesis. These are exactly the features on which methylation detection relies, and their absence in PBSIM3 output is the gap KinSim is designed to fill. The Sequel-vs-Revio chemistry mismatch matters little here: read-length distributions and post-CCS HiFi quality are comparable, and the kinetic signatures themselves are a property of the φ29 polymerase's response to modified bases, not of the chemistry version. A natural follow-up would be to recalibrate PBSIM3's length and error profiles on empirical Revio data, but this is orthogonal to the kinetic-injection problem addressed here.

3.4.4 Allignement Tools

1. Minimap2

Minimap2 [36] is a versatile pairwise alignment tool for long reads, designed to map noisy long reads (PacBio CLR, ONT) and assembled contigs to a reference genome. It uses a minimizer-based seeding strategy combined with chaining and base-level dynamic programming to produce accurate split-read alignments efficiently, handling sequencing error rates of up to 15% (Li, 2018). Minimap2 supports multiple alignment modes optimized for specific data types, including HiFi reads (map-hifi), CLR/ONT noisy reads (map-pb and map-ont), splice-aware RNA-seq alignment, and assembly-to-reference alignment. Its output follows the standard SAM/PAF formats, making it compatible with downstream tools in the standard bioinformatics ecosystem. Minimap2 is now the de facto standard for long-read alignment, used in the vast majority of PacBio and ONT analysis pipelines, and significantly outperforms previous tools (BLASR, BWA-MEM long-read mode) in both speed and accuracy on long-read data (Li, 2018).

2. Pbmm2

pbmm2 (PacBio's official wrapper around Minimap2) [37] is a SMRT-compatible alignment tool that integrates Minimap2 with PacBio-specific data handling. While Minimap2 itself can align PacBio reads via the map-pb or map-hifi modes, pbmm2 ensures full compatibility with the PacBio BAM format and its associated metadata (including the kinetic tags fi, fp, ri, rp, ip, pw that encode IPD and PW per base). The wrapper preserves these PacBio-specific tags during alignment, applies sensible default parameters tuned for PacBio data, and handles indexed reference genomes via the .mmi format for efficient repeated alignments. pbmm2 is the recommended alignment tool when downstream analyses require access to the original kinetic information: for example, methylation detection via tools such as ipdSummary, modkit, or fibertools-rs. For standard sequence-only analyses, Minimap2 directly is sufficient, but for kinetic-aware workflows, pbmm2 is the appropriate choice.

4 Methydology and Implementation

4.1 Methodology

All computational work was performed on the IBU high-performance computing cluster at the University of Bern, accessed through the BeFri partnership between the Universities of Fribourg and Bern. The cluster is managed by the SLURM workload scheduler and provides access to thousands of CPU cores, GPU-equipped nodes, and up to 950 GB of RAM per node. Data preparation, extraction, and refinement were run on CPU nodes, while model training was performed on GPU nodes equipped with NVIDIA RTX 2080 Ti cards.

Software environment management combined two approaches. PacBio SMRT-Link tools (ipdSummary, pbmotifmaker, Jasmine, pbmm2) were executed via Apptainer containers or a specific install on the cluster, allowing access to the most up-to-date versions of these tools without relying on cluster-wide installations. For the Python-based components of KinSim, a Conda environment was used, providing Python 3.9, PyTorch 2.6.0 with CUDA 12.4 support, pysam, numpy, scikit-learn, and plotly.

The KinSim codebase is maintained as a public Git repository on GitHub, with a single command-line entry point (kinsim) exposing all pipeline stages: extract, refine, train, generate, evaluate, and analyze. Configuration is centralised in a YAML file that defines the methylation alphabet, signature offsets, model hyperparameters, and refinement parameters. The repository also documents the exact versions of all external dependencies and tools used, ensuring that any experiment can be fully reproduced by providing the same configuration file and input data

4.2 Development approach

The development of KinSim followed an iterative approach, starting from the simplest possible solution and increasing complexity only when justified by empirical evidence.

The first attempt was inspired by the logic of ipdSummary: build a baseline model of unmodified kinetics, then identify deviations caused by methylation. However, unlike ipdSummary, no whole-genome amplified (WGA) data was available to serve as an unmodified reference. Without WGA, the baseline had to be estimated directly from the native sequencing data, where methylated and unmethylated signals are mixed. The initial approach was deliberately simple: for each methylation type (e.g. m6A, which modifies adenines), every adenine in the genome was treated as a potential observation. Since the vast majority of adenines in a bacterial genome are not methylated, the aggregate IPD distribution across all A positions should approximate a baseline, and methylated sites should appear as outliers. This approach failed because the sequence context surrounding each base dominates the IPD signal: an adenine in AAAAAAA produces a very different IPD than an adenine in GCGAGCG, and this context-dependent variation completely masks the subtle methylation effect.

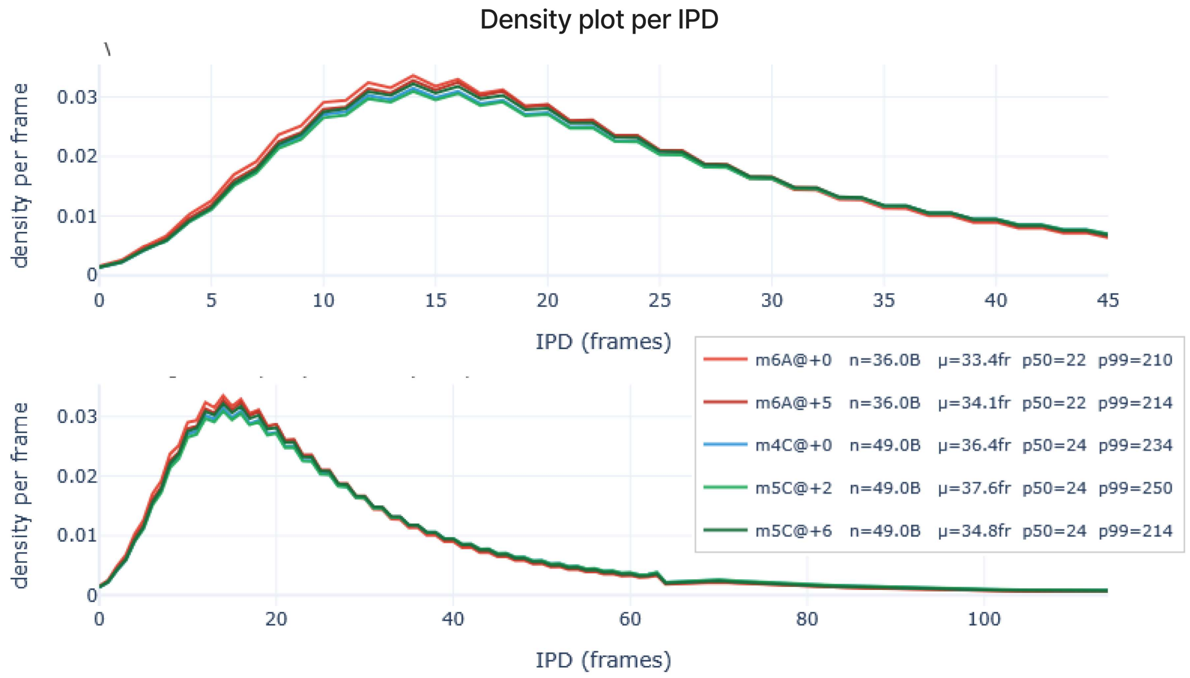


Figure 14: Density plot IPD

Figure 14 confirms this: it shows the per-frame IPD density for positions affected by each methylation type. The distributions for methylated positions (With the main ipd Ratios of m6A at offsets 0 and +5, m4C at offset 0, m5C at offsets +2 and +6) are nearly indistinguishable from the global baseline when plotted without sequence context normalisation. Only when examining the same k-mer with and without methylation does the shift become visible. This observation established that any predictive model must be k-mer-aware.

The first k-mer-aware architecture was a flat k-mer embedding table (MLPPredictor), which assigned a separate learned vector to each of the 4^{11} = 4.2 million possible 11-mers. While this model could memorise per-k-mer distributions, it had 268 million parameters, 99.98% of which resided in the embedding table, and could not generalise across similar k-mers or transfer knowledge between methylation states. Given the 95/5 class imbalance between unmethylated and methylated samples, this was a critical limitation: the model saw too few methylated examples per k-mer to learn a meaningful methylation effect.

The second iteration, ConvPredictor, addressed both shortcomings of the first. Rather than learning a separate embedding per k-mer, it embeds each base individually, four entries instead of 4.2 million, and processes the resulting sequence through one-dimensional convolutions, which learn compositional rules from the local context rather than memorising each k-mer in isolation. Methylation information is then injected through FiLM conditioning, applying a learned per-channel scale and shift derived from the methylation state at each position. This factorisation lets the model share its representation of what methylation does to kinetics across all sequence contexts, a property the per-k-mer table fundamentally could not have. The architecture is described in detail in Section 4.3; with the default configuration the total parameter count was approximately 140 000, low enough to force the model to generalise rather than memorise. The window geometry was inherited from earlier per-position kinetics work, with seven bases upstream and three downstream of the active site.

ConvPredictor was the first version of the system to be trained and evaluated end to end, and it confirmed that the broad strategy, predicting per-base kinetics from sequence and methylation context, was viable. It also surfaced four limitations of the design that progressively motivated a complete redesign.

The first concerns the output formulation. ConvPredictor predicts a Gaussian mean and standard deviation per position, trained under Gaussian negative log-likelihood. This is a clean and well-calibrated parametric family for log-quantised PacBio kinetics, but the log1p transform does not make the underlying distribution exactly Gaussian; it merely makes it close enough for a first approximation.

The second concerns the extraction strategy. The training data was extracted only at positions matching motifs declared in the strain's motifs.csv, supplemented by a small baseline pool. As a consequence, the model received a strongly motif-anchored view of the genome: well-represented motifs were over-sampled, motifs absent from the catalogue were unseen, and the distinction between "this position lies on a known motif but its methylation did not fire" and "this position is genuinely unmethylated" was never made explicit in the training signal. Downstream interpretation therefore required a separate post-processing step to disentangle the contributions of motif presence, motif firing rate, and sequence context. [Figures: bias introduced by motif-anchored extraction.]

The third concerns the prediction granularity. ConvPredictor predicts a single position at a time on the forward strand, with only the immediate ±1 bp visible to the model on the reverse strand. This is doubly limiting. At generation time, each emitted base requires its own forward pass, making generation roughly two orders of magnitude slower than the downstream alignment and methylation-calling stages it feeds into. And, more importantly, the model has no representation of the joint kinetic profile across positions, which is precisely the signature that distinguishes a true methylation event from a sequence-context artefact.

The fourth limitation follows directly from the third. Because the model never produces a joint prediction over a window, it cannot reconstruct the shape of the kinetic signature around a methylated base. Sampling from ConvPredictor at each position independently and concatenating the results does not recover the position-to-position correlations that a real polymerase produces: the marginals can be correct while the joint shape is wrong. [Figures: empirical IPD/PW envelopes around m6A sites in real data versus per-position ConvPredictor samples.]

These four points, taken together, also explain why the natural next step was not to scale up the convolutional architecture. A wider CNN, or one with a deeper receptive field, would still produce per-position outputs trained against a unimodal parametric likelihood. The limitations of ConvPredictor are not limitations of model capacity; they are limitations of the output formulation (parametric, per-position) and of the training-data definition (motif-anchored, marginalised over a single position). A productive redesign therefore had to revisit both at once.

The redesign was framed as two separate decisions taken in sequence: the training objective first, and then the architectural backbone. The first decision was to drop the parametric likelihood and train the model adversarially against its own samples, using the Wasserstein objective with gradient penalty (Gulrajani 2017). This addresses the output-formulation problem directly — there is no Gaussian assumption, no log-skew approximation, and the joint distribution over the kinetic window is whatever the data require, and lifts the per-position factorisation by working sample-by-sample on whole windows rather than position-by-position on marginals.

The second decision concerns the backbone for the generator and the critic, once the choice of an adversarial sample-level objective was settled. On a 21-position (Similar to Jasmine's) window the receptive-field argument that motivates deep CNNs in genomics, covering long-range dependencies, does not apply: a handful of dilated convolutions, or self-attention at $K^2 = 441$ operations per token, both span the entire window cheaply, and neither dominates the other on compute. The decisive criterion was therefore the *conditioning mechanism*, not the receptive field. AdaLN-Zero (Peebles & Xie, "Scalable Diffusion Models with Transformers", ICCV, 2023, doi:10.48550/arXiv.2212.09748) replaces the affine parameters of every layer normalization by a per-sample (shift, scale, gate) triple derived from the conditioning vector, with the gate zero-initialised so each block starts as the identity transform. This zero-initialisation property empirically smooths the early phase of training for conditional generators and has no clean analogue under FiLM-on-CNN, where the conditioning enters only as a per-channel affine on the convolutional features and the network does not start as identity. The same family of blocks is used for the critic, which keeps the spectral-norm constraint that supports the 1-Lipschitz target uniformly across the architecture, with manually unrolled attention so the double backward required by the gradient penalty remains tractable, a property that the fused fast-attention kernels available in PyTorch do not currently provide. The transformer also makes the joint structure of the 21×4 output explicit through attention rather than implicit through stacked convolutions, which proved useful when diagnosing per-bucket failures during training. A 1-D CNN cGAN remains a reasonable alternative at this window size and was not ruled out on grounds of expected performance, but on the cleanness of conditioning and the transparency of the joint-structure modelling.

With these two choices made, the resulting architecture takes the following form. The generator receives a 21-position context, the per-position base identities on both strands and the per-position methylation labels on both strands together with a latent noise vector, and emits the full 21-position by 4-channel kinetic tile (IPD and PW on the forward and reverse strand) in a single forward pass. Both the generator and the Critic see the same conditioning context, which is what makes the model genuinely conditional rather than an unconditional sampler with side information.

This design resolves the four limitations of ConvPredictor by construction. First, the output is non-parametric: the joint distribution over the kinetic tile is whatever the data require, learned implicitly through the adversarial objective. Second, the input is taken directly from the aligned BAM rather than from motif-anchored excerpts: positions are drawn from reads with no motif catalogue intermediating, which removes the catalogue-coverage bias entirely and allows the same model to be applied to organisms for which no motif catalogue is available. Third, generation is one forward pass per window, every position in the window is emitted simultaneously — so the per-base inference cost is reduced by roughly an order of magnitude relative to ConvPredictor. Fourth, the joint output over the window means the position-to-position correlations within the kinetic signature are modelled directly, and the characteristic shape of the IPD response around a methylation centre is reproduced as a single coherent draw rather than as a sequence of independent ones.

The architecture is correspondingly larger, approximately 12 M parameters in the generator and 3 M in the critic, against roughly 140 000 for ConvPredictor — but the increase is proportional to the broader task. A 21-by-4 joint output conditioned on a 21-position bilateral context is fundamentally a richer prediction than a single position's (μ, σ) pair, and the additional capacity is required to represent it without underfitting. The complete architecture, training procedure, and design choices are detailed in Section 4.3.

4.3 Methods

4.3.1 Data extraction

The aim of the extraction stage is to convert a corpus of real aligned PacBio HiFi BAMs into a tabular training set in which each row pairs a local *sequence + methylation context* with the *per-strand kinetic measurements* that a polymerase actually produced at that locus. Because methylation kinetics depend on both the local sequence and the type of modification present *within a window* around the focal position, the extraction must record enough context on both strands to allow the model to recover this joint dependence.

- **Input data**

Extraction operates on aligned, bystrandified BAM files (one /ccs/fwd and one /ccs/rev record per ZMW, carrying the per-strand ip (Inter-Pulse Duration) and pw (Pulse Width) kinetic tags), aligned to the strain's reference assembly with pbmm2. A separate methylation annotation file, a motifs.gff produced upstream by ipdSummary and pbmotifmaker (Pacific Biosciences, same reference) and filtered to motif-confirmed positions with modificationQv ≥ 10, provides the genomic positions and types of methylation present in the strain.

- **Window geometry**

For every focal position retained for extraction, a window of K = 21 reference bases is built, centred on the focal position (positions −10 to +10). Each row of the resulting shard contains: the four kinetic channels at all 21 positions (IPD_fwd, PW_fwd, IPD_rev, PW_rev), stored as uint8 according to the PacBio CodecV1 frame-count; the methylation identifier on the forward strand at every position; the methylation identifier on the reverse strand at every position (the redesigned extraction stores the full K-length methylation context on both strands, lifting the ±1 reverse-strand restriction used by ConvPredictor, palindromic motifs are not the only modification class encountered in the corpus, and the slowing signature extends to offsets well beyond ±1); the strain identifier and the focal genomic position, used for stratified train/test splits; and a category label (baseline, slowed, near_meth).

- **Three-category labelling**

The dominant difficulty in PacBio kinetic data is that the kinetic effect of a methylation is *not* localised on the modified base itself: m6A produces an additional IPD peak at offset +5, m5C at offsets +2 and +6, m4C only at offset 0 (Feng et al., *PLoS Computational Biology* 9(3):e1002935, 2013, cited above). Treating only the modified base as the methylation site discards the very signal downstream callers rely on. Extraction therefore emits, for every annotated methylation position m of type T, *multiple* rows at offsets +k from m:

if k ∈ signal_offsets[T], the row is labelled slowed, the focal position falls on a position where the polymerase is expected to deviate a lot from baseline kinetics

if k ∈ [+1, +10] but k ∉ signal_offsets[T], the row is labelled near_meth, close to a methylation but at an offset where no kinetic perturbation is expected (a negative-control class);

positions further than 20 bp from any annotated methylation are eligible for the baseline pool. A streaming reservoir samples baseline_per_strain such positions per strain. All this is done to balance training.

- **Strand-aware kinetic routing**

A read whose is_reverse flag is set has its base sequence reverse-complemented by the aligner, but the bystrandify + pbmm2 toolchain (Pacific Biosciences, *SMRT Tools User Guide v13.1*, 2024) does *not* reverse the per-base kinetic tag arrays accordingly. Extraction therefore reverses the ip and pw arrays internally whenever the BAM-stored sequence and the kinetic tag have opposite orientations relative to the original CCS read (a property determined jointly by the record suffix /fwd vs /rev and the is_reverse flag). Without this correction, roughly half of the extracted samples would carry kinetics measured at the wrong query position.

- **Caps and balancing**

Two caps bound BAM I/O and training-time imbalance: a *parent methylation cap* (meth_per_strain_cap, default 15 000 positions per strain) and a *per-position read cap* (reads_cap_per_position, default 20).

- **Output**

For the 65 strains of the present corpus (52 *Streptomyces* + 13 Vega isolates, all sequenced on the PacBio Revio platform with P1-C1 chemistry), the resulting shards total approximately 50 GB.

4.3.2 Model

The generative task is fundamentally distributional: at each (sequence context, methylation context) configuration, polymerase kinetics produce not a single value but a distribution of IPD and PW measurements that the model must reproduce. KinSim addresses this through a conditional Generative Adversarial Network, trained under the Wasserstein formulation with gradient penalty. This choice over a Gaussian-likelihood regression is motivated by three properties of the data: per-context distributions in PacBio uint8 codec space are heavy-tailed and not always unimodal; inter-read variance dominates the mean shift at most positions, making a sample-level adversarial loss more informative than a per-sample mean-squared error; and the WGAN-GP objective supplies a meaningful training gradient even when the generator's output distribution is far from the real distribution, avoiding the mode-collapse failure modes of earlier GAN formulations.

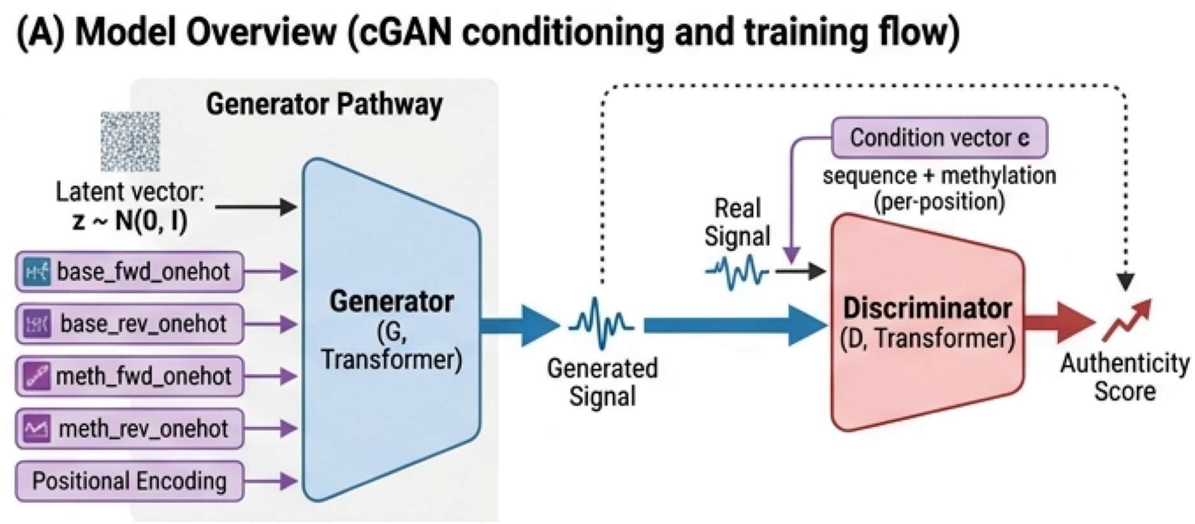
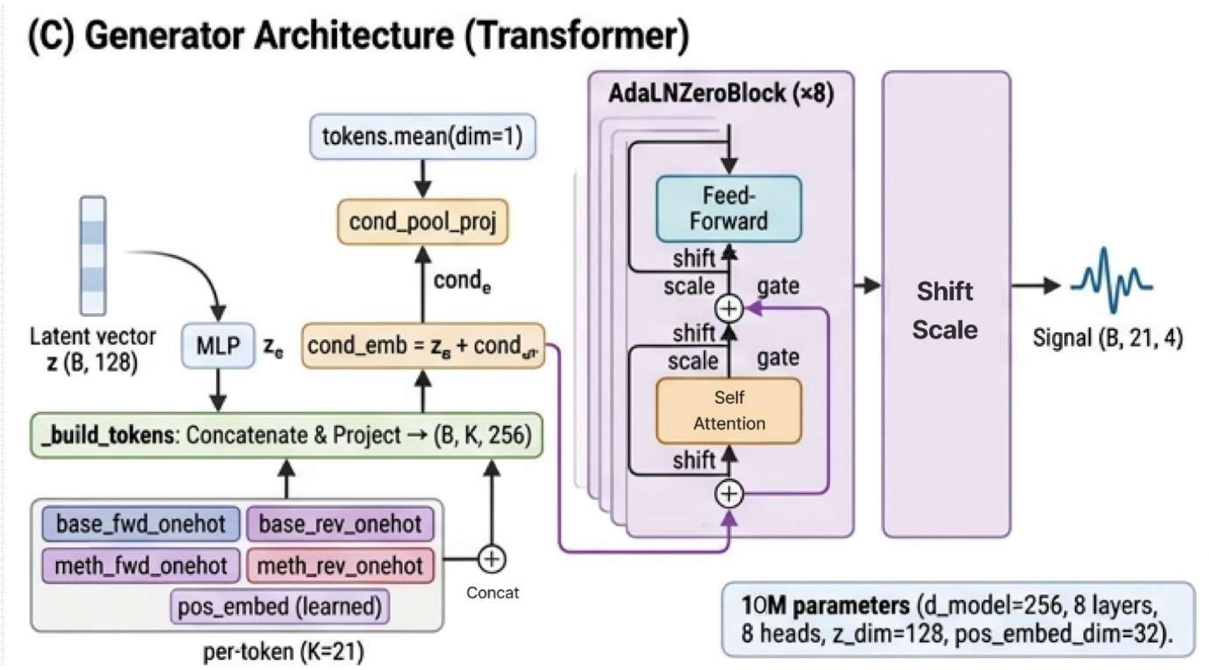


Figure 15: cGAN principles schema

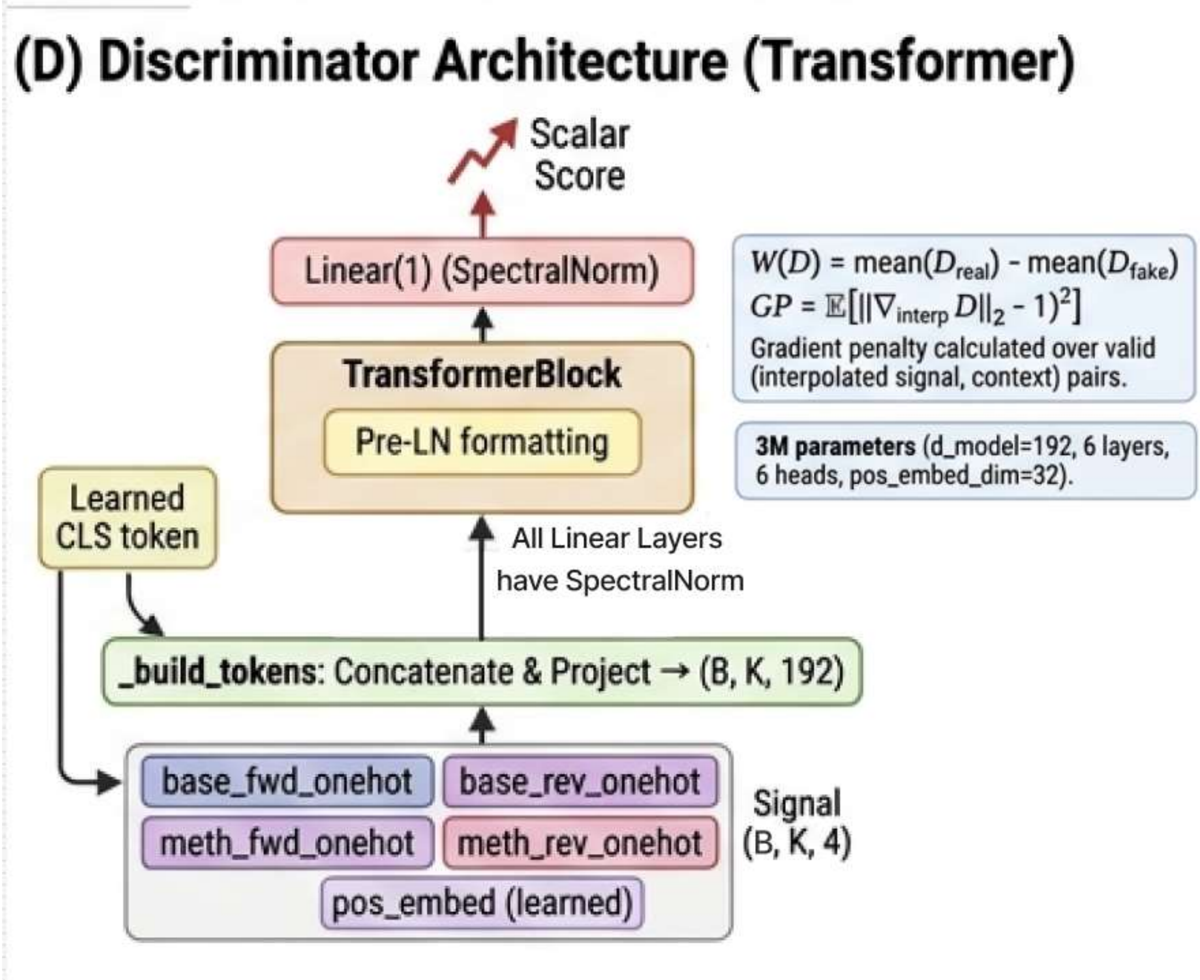
Generator

The generator is a Transformer with eight AdaLN-Zero blocks with d_model = 256, eight attention heads and mlp_ratio = 4.0, totalling approximately 12 M parameters. The input layer concatenates, at every position of the 21-base window: the one-hot forward base, the one-hot reverse-complement base, the one-hot methylation identifier on the forward strand, the one-hot methylation identifier on the reverse strand, and a learned positional embedding of dimension 32. A per-sample latent vector $z \in \mathbb{R}^{128}$ (sampled from $\mathcal{N}(0, I)$ at every training step) carries the stochasticity needed to generate distributions rather than point estimates. z is mapped through a small MLP and combined with a mean-pooled summary of the conditioning tokens, producing a per-sample conditioning vector broadcast into every AdaLN-Zero block as the modulation source. The zero-initialised gate inside AdaLN-Zero ensures the network starts as a near-identity mapping at initialisation, a property shown to substantially stabilise the early phase of generative training in Peebles & Xie (ICCV, 2023, cited above). The output head, also zero-initialised, projects the final hidden states to four real-valued channels per position (IPD_fwd, PW_fwd, IPD_rev, PW_rev), trained in log1p(frames) space and decoded to PacBio CodecV1 uint8 at generation time.



Critic

The Critic (the WGAN-GP analogue of a classical discriminator) is a smaller Transformer (d_model = 192, six layers, six heads, ~3 M parameters). As illustrated on Figure 15, it receives the same conditioning tokens as the generator, concatenated with the candidate signal, either a real shard sample or a generated one, and predicts a single scalar score per sample. The critic enforces 1-Lipschitz behaviour through a two-sided gradient penalty applied on interpolated samples (Gulrajani et al., NeurIPS, 2017, cited above); spectral normalisation (Miyato et al., "Spectral Normalization for Generative Adversarial Networks", International Conference on Learning Representations (ICLR), 2018, doi:10.48550/arXiv.1802.05957) is additionally applied to every linear projection inside the critic to further regularise its Lipschitz constant.



Loss and conditioning mechanism

The Critic is trained to maximise $\mathbb{E}[D(\text{real})] - \mathbb{E}[D(\text{fake})] - \lambda \cdot \text{GP}$, and the generator to maximise $\mathbb{E}[D(\text{fake})]$, with $\lambda = 10$. The critic is updated n_critic = 5 times for every single generator update following the two-time-scale update rule [38], with learning rates 4×10^{-4} for the critic and 1×10^{-4} for the generator. The methylation information acts on the generator at two complementary levels. First, it appears directly in the input tokens as a one-hot identifier per position, allowing the attention layers to route information based on local methylation pattern. Second, the mean-pooled conditioning summary feeds into the AdaLN-Zero modulation of every block, which is mathematically equivalent to a per-channel FiLM affine transformation applied after layer normalization, letting the methylation context globally modulate the magnitude and offset of every feature channel.

4.3.3 Evaluation Metrics

Evaluating a methylation-conditioned generative model requires assessing the data at two distinct levels: distributional fidelity at the sample level, does the generated joint distribution of (context, meth, kinetic) triples match the real one? And end-to-end functional fidelity, does running the standard methylation-calling pipeline on a generated BAM recover the same motifs as on a real BAM?

Per-category Wasserstein-1 distance. At training time, the model is evaluated every 5 000 steps on a held-out set of strains never seen during training (test_strains = {bc2034, bc2045, bc2048, bc2082}). For each evaluation, generated samples are produced for all (context, meth) combinations present in the held-out shards, and the empirical 1-D Wasserstein distance W_1 is computed between the generated and real distributions of the central kinetic channel, bucketed two ways. The first bucketing is per methylation type (baseline / m6A / m4C / m5C), exposing whether the model has learned the correct marginal shape of each class. The second is per emission category (baseline / slowed / near_meth), exposing whether the model has learned the offset structure of the kinetic effect, as characterised by Feng et al. (PLoS Computational Biology 9(3):e1002935, 2013, cited above) — a generator that only produces a single mean shift at the methylation centre but ignores the +5 peak for m6A would have a low W_1 on the centre bucket but a high one on the offset bucket. A best-so-far checkpoint is saved whenever the overall W_1 reaches a new minimum, and its trajectory across training is reported in Section 5.

Training-time diagnostics. Three quantities are logged every 100 steps and reported in Section 5: (i) the critic loss $D = \mathbb{E}[D(\text{real})] - \mathbb{E}[D(\text{fake})]$, whose magnitude is the Wasserstein-distance estimate that the WGAN-GP objective optimises (Gulrajani et al., NeurIPS, 2017, cited above); (ii) the gradient-penalty term GP, which should remain small (around 10^{-2} in well-behaved runs) — a growing GP indicates that the critic is violating the Lipschitz constraint, often a precursor to mode collapse; (iii) the per-batch generator score $G = \mathbb{E}[D(\text{fake})]$, whose oscillation pattern reflects the equilibrium between generator and critic.

Byte-level cross-check of the extraction pipeline. Before any training-time evaluation, the extraction pipeline itself is cross-validated against the raw HiFi BAMs from which the bystrandified BAMs originate. For a sample of strong motif-confirmed methylation positions (IPDRatio $\geq$ 5, modificationQv $\geq$ 100, as reported by ipdSummary per the SMRT Tools User Guide v13.1, Pacific Biosciences, 2024), the kinetic byte values stored in the shard are compared, ZMW-by-ZMW, against the values obtained by reading the raw fi and ri tag at the corresponding query position. This test identified and confirmed the strand-aware tag reversal bug documented in Section 4.3.1: after correction, the shard-vs-raw match reaches 100 % on 18 reads at three independently chosen positions, and within sampling noise across 4 600 reads at 50 positions.

End-to-end motif recovery. The ultimate functional test is to run the full upstream methylation-calling pipeline on a BAM generated by KinSim and to compare the discovered motifs against those discovered on the real BAM of the same strain. For each of the four held-out strains, the generated BAM is processed through ccs-kinetics-bystrandify, pbmm2 align, ipdSummary, and pbmotifmaker (all from Pacific Biosciences, SMRT Tools User Guide v13.1, 2024, cited above; pbmotifmaker run with --min-score 40), in parallel with the Jasmine 5-methylcytosine caller (Tse et al., "Genome-wide detection of cytosine methylation by single molecule real-time sequencing", Proceedings of the National Academy of Sciences 118(5):e2019768118, 2021, doi:10.1073/pnas.2019768118) and modkit (Oxford Nanopore Technologies, modkit v0.2, 2024, https://github.com/nanoporetech/modkit). The resulting motif lists SIM_motifs_merged.csv and real_motifs_merged.csv are compared by motif string at a fraction threshold of 0.7. A true positive is a motif present in both lists; a false positive a motif present only in the simulation; a false negative a motif present only in the real data. Precision, Recall, and F1 score are reported per strain in Section 5.

Baseline comparison. Two baselines anchor the interpretation of the cGAN results: a naive Gaussian baseline, one global $\mathcal{N}(\mu\_T, \sigma^2\_T)$ per methylation type T, ignoring sequence context entirely, that the model must outperform to demonstrate that conditioning on sequence has been learned; and a per-context oracle, the empirical mean and standard deviation of the real data at every observed (k-mer, methylation) combination, with no generalization, the theoretical ceiling for any per-context predictor. The cGAN's W_1 is expected to lie strictly between these two bounds; the fraction of the gap it closes quantifies what proportion of the achievable distributional fidelity the model has captured.

5 Results

5.1 Training corpus

The training data consists of three sources of real PacBio HiFi sequencing reads, all encoded with the per-base kinetic tags fi, fp, ri, rp and aligned to per-strain reference genomes with pbmm2.

After quality filtering, the retained corpus comprises:

- **52 Streptomyces strains**, PacBio Revio platform, R/P1-C1 chemistry (used). This dataset was provided by Dr. L. Falquet's group at the University of Fribourg.

- **13 bacterial** isolates from a single Vega instrument run, R/P1-C1 chemistry (used). This dataset was contributed by D. Stucki (Pacific Biosciences) in April 2026 specifically to broaden the training distribution to the new Vega chemistry and instrument generation.

MSA-1003 mock-community sample, PacBio Revio, R/P1-C1 chemistry (unused). This 20-organism mock community was identified as a candidate for downstream functional validation of the methylation-aware binner

5.2 Pipeline validation

Before any model-quality evaluation could be conducted, the extraction and emission halves of the pipeline had to be validated end to end. PacBio long-read kinetics are stored and processed under a set of conventions that are individually well documented but whose interactions, particularly across the bystrandify, pbmm2 alignment, and ipdSummary stages, are easy to mis-read. Validation therefore consisted of two activities: a per-position cross-check between the extracted training tensors and the raw aligned BAM, and an end-to-end pass of the generated BAM through the downstream methylation-calling chain.

5.3 Distributional fidelity

[On its way]

5.4 End-to-end motif recovery

[Getting Generated]

6 Conclusion

Not yet

6.1 Summary of contributions

Not yet

6.2 Limitations

Not yet

6.3 Future work

Not yet

6.4 Closing remarks

Not yet

Glossary

BAM Binary Alignment/Map

BIC Bayesian Information Criterion

BiGRU Bidirectional Gated Recurrent Unit

BiLSTM Bidirectional Long Short-Term Memory

BPE Byte Pair Encoding

CCS Circular Consensus Sequencing

cGAN Conditional Generative Adversarial Network

CLR Continuous Long Read

CNN Convolutional Neural Network

CRF Conditional Random Field

FiLM Feature-wise Linear Modulation

GAN Generative Adversarial Network

GBM Gradient Boosted Machine / Regression Tree

GMM Gaussian Mixture Model

GRU Gated Recurrent Unit

HiFi High Fidelity

HMM Hidden Markov Model

IPD Inter-Pulse Duration

PW Pulse Width

KDE Kernel Density Estimation

LSTM Long Short-Term Memory

m4C (4mC) N^4-methylcytosine

m5C (5mC) 5-methylcytosine

m6A (6mA) N^6-methyladenine

MAG Metagenome-Assembled Genome

MDN Mixture Density Network

MLP Multi-Layer Perceptron

MSE Mean Squared Error

MTase Methyltransferase

NLL Negative Log-Likelihood

ONT Oxford Nanopore Technologies

PW Pulse Width

REase Restriction Endonuclease

R-M Restriction-Modification

RNN Recurrent Neural Network

SMRT Single Molecule, Real-Time

TNF Tetranucleotide Frequency

VAE Variational Autoencoder

WGA Whole-Genome Amplified

WGAN-GP Wasserstein GAN with Gradient Penalty

ZMW Zero-Mode WaveguideWaveguide

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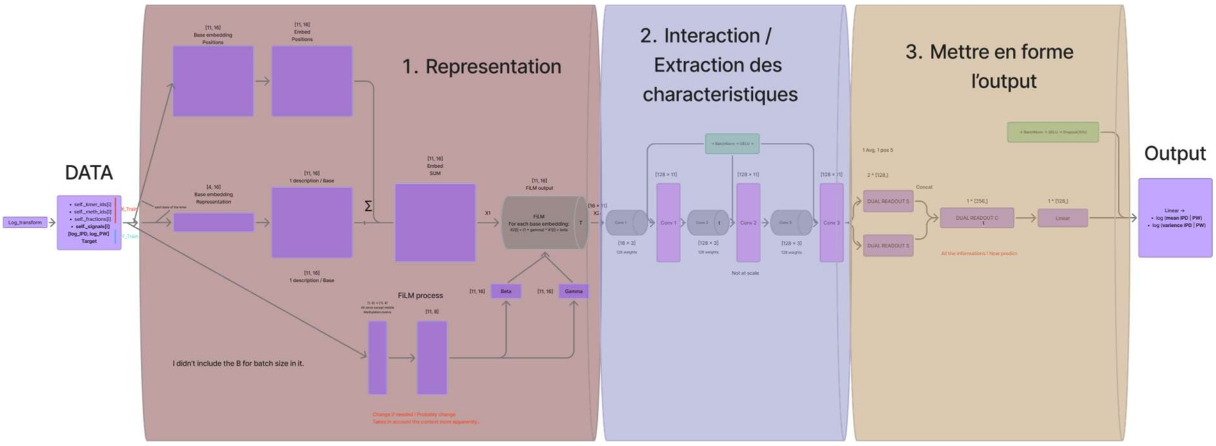
Poster

Appendices

A. Representations

AI Convolution model architecture



B. Github

https://github.com/NicolasDutilleux/KinSim.git